On the Spectrum of Path-Length Density Matrices and its Connections with the Topology of Unrooted Binary Trees

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Abstract

An *Unrooted Binary Tree* (UBT) T is a tree with $n \geq 3$ leaves and internal nodes of degree 3. The *Path-Length Density Matrix* (PLDM) of a UBT T is a symmetric matrix of order n having zero diagonal entries and generic nondiagonal entry equal to $2^{-\tau_{ij}}$, where τ_{ij} is the number of edges on the unique path in T between leaves i and j. This particular class of matrices finds a fundamental application in coding and information theory as well as some highly nonlinear $\mathcal{NP}$-hard network design problems that arise from the life sciences. In this article, we investigate the extent to which the spectrum of a PLDM reflects the topology of the underlying UBT. We start by completely characterizing the spectrum of PLDMs for specific trees, and we subsequently show that rational eigenvalues of a PLDM reflect the presence (or absence) of specific subtrees in the associated UBT. Then, we present a recursive formula to compute the eigenvalues of the PLDM of an arbitrary UBT and show extremal properties of the eigenvalues of PLDMs. As a byproduct, we shed a new light on the extremal structure of the spectrum of Kac-Murdock-Szegö matrices. Finally, we conclude by describing open problems on the spectral conditions that must be satisfied by a symmetric matrix to encode PLDMs of UBTs.

Keywords: combinatorial matrix theory, algebraic graph theory, unrooted binary trees, tree realization, path-length matrices, extremal combinatorics

1. Introduction

Terminal encodings are matrix-based representations of tree graphs that depend exclusively on information associated with a distinguished subset of nodes, typically referred to as *terminals*. Such encodings arise naturally in inverse optimization problems that involve reconstructing or identifying an optimal tree structure using only incomplete or aggregated information, often available only at terminal nodes. Classical examples include the *Huffman Coding Problem* (HCP) (Parker and Ram, 1999) and the *Balanced Minimum Evolution Problem* (BMEP) (Catanzaro et al., 2025), where the searched tree structure

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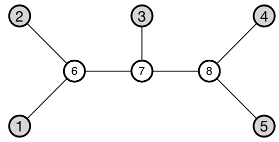

$$\tau = \begin{pmatrix} 0 & 2 & 3 & 4 & 4 \\ 2 & 0 & 3 & 4 & 4 \\ 3 & 3 & 0 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 4 & 4 & 3 & 2 & 0 \end{pmatrix} \qquad \mathbf{2}^{-\tau} = \begin{pmatrix} 0 & 1/4 & 1/8 & 1/16 & 1/16 \\ 1/4 & 0 & 1/8 & 1/16 & 1/16 \\ 1/8 & 1/8 & 0 & 1/8 & 1/8 \\ 1/16 & 1/16 & 1/8 & 0 & 1/4 \\ 1/16 & 1/16 & 1/8 & 1/4 & 0 \end{pmatrix}$$

Figure 1: On the left, an example of an Unrooted Binary Tree (UBT), i.e., a tree graph having $n \geq 3$ leaves, $n-2$ internal nodes each of degree 3, and $2n-3$ edges. The UBT shown in this example has 5 leaves, 3 internal nodes, each of degree 3, and 7 edges. At the center and on the right, the Path-Length Matrix (PLM) $\boldsymbol{\tau}$ and the Path-Length Density Matrix (PLDM) $\mathbf{2}^{-\boldsymbol{\tau}}$ induced by this UBT, respectively.

is inferred from pairwise relationships between terminals (expressed, e.g., in terms of symbol frequencies in the case of the HCP, or evolutionary distances in the case of the BMEP) by optimizing an information theoretic objective function (such as the Shannon entropy of the terminal distribution in the case of the HCP, or the cross-entropy of the terminal distribution in the case of the BMEP Catanzaro et al. (2020a)). A prominent example of a terminal encoding, particularly relevant when the set of terminals coincides with the set of leaves of a tree T, is the *Path-Length Matrix* (PLM) (Catanzaro et al., 2025). Given a tree T with $n \geq 3$ leaves, the PLM is an integer square matrix $\tau = \{\tau_{ij}\}$ of order n, whose generic entry τ_{ij} represents the number of edges on the unique path in T connecting leaves i and j. A closely related form of terminal encoding, which plays a central role in this article, is the *Path-Length Density Matrix* (PLDM). PLDMs are particularly useful when (i) the set of terminals coincides with the set of leaves of T, and (ii) the internal nodes of T are subject to strict degree constraints. Under assumption (i), the PLDM induced by a tree T whose internal nodes all have degree exactly δ is a real square matrix of order n, with zero diagonal entries and off-diagonal entries equal to $(\delta-1)^{-\tau_{ij}}$. In this article, we study the relationships between the spectral properties of PLDMs and the structures of the associated trees, particularly when these trees are *unrooted* and *binary* (see, e.g., Figure 1). Spectral methods are known to exhibit intrinsic limitations in characterizing tree topologies: in general, unweighted trees are not uniquely determined by the spectrum of their adjacency matrix (Schwenk, 1973) nor by that of their distance matrix (McKay, 1977), and analogous non-uniqueness phenomena also arise for rooted binary trees (Matsen and Evans, 2012). Nevertheless, Schwenk showed that cospectrality in trees is often induced by the presence of specific rooted subtrees that can be interchanged without affecting the spectrum. This structural observation suggests that cospectral trees may share a common core carrying identical spectral information, leaving open the possibility of recovering meaningful portions of the underlying topology from spectral data . Here, we show that, within the framework of terminal encodings, the spectrum of a PLDM encodes a substantial amount of information about the structure of its associated unrooted binary tree (UBT). After introducing in Section 2 some notation, definitions, and preliminary results, we characterize in Section 3.2 the spectra of PLDMs associated with UBTs attaining the minimum and maximum possible diameters. In the same section, we derive a recursive formula for computing the eigenvalues of the PLDM of an arbitrary UBT and establish several extremal spectral properties. In Section 3.1, we show that the presence or absence of rational eigenvalues in a PLDM reflects the occurrence of specific subtree configurations in the underlying UBT. In Section 3.3, we strengthen these results by further analyzing rational eigenvalues and by studying the asymptotic behavior of the PLDM spectrum. As a byproduct, this analysis yields new insights into the extremal spectral structure of Kac–Murdock–Szegő matrices. In Section 3.4, we establish a relationship between the diameter of a UBT and the spectrum of its associated PLDM. Finally, in Section 4, we outline several open problems concerning the spectral conditions that a symmetric matrix must satisfy in order to represent the PLDM of a UBT.

2. Notation and preliminaries

Vectors and matrices. Given two integers a, b with $a \leq b$, we denote by $[a, b]$ the set of integers c such that $a \leq c \leq b$. When $a = 1$, we simply write $[b]$. For any integer $n \geq 1$ and any $i \in [n]$, we denote by $\mathbf{e}_i$ the i-th canonical unit vector of $\mathbb{R}^n$. We also denote by $\mathbf{0}_n$ and $\mathbf{1}_n$ the column vectors in $\mathbb{R}^n$ whose entries are all equal to 0 and 1, respectively, and by $\mathbf{I}_n$ the identity matrix of order n. Unless stated otherwise, any matrix $\mathbf{M}$ is assumed to be real, symmetric, and of order n. We denote the determinant, rank, kernel, and image of a matrix $\mathbf{M}$ by $\det(\mathbf{M})$, $\operatorname{rank}(\mathbf{M})$, $\ker(\mathbf{M})$, and $\operatorname{im}(\mathbf{M})$, respectively, and we recall that the rank and the kernel are related by the following fundamental result from linear algebra (Horn and Johnson, 2013):

Theorem 1 (Rank-nullity theorem)**.**

$$\operatorname{rank}(\mathbf{M}) + \dim(\ker(\mathbf{M})) = n.$$

We refer to the i-th row and column vectors of $\mathbf{M}$ as $\mathbf{M}_{i\cdot}$ and $\mathbf{M}_{\cdot i}$, respectively, and we denote by $\operatorname{diag}(m_{11}, m_{22}, \ldots, m_{nn})$ the canonical diagonal matrix having m_{11}, m_{22}, …, m_{nn} as diagonal entries. Throughout the article, we shall use lowercase letters for scalars (e.g., $c \in \mathbb{R}$), bold capital letters for matrices (e.g., $\mathbf{M} \in \mathbb{R}^{n\times n}$), bold lowercase letters for vectors (e.g., $\mathbf{v} \in \mathbb{R}^n$), and Greek letters to denote the eigenvalues of a given matrix. In addition, given a matrix $\mathbf{M}$, we shall refer to the algebraic multiplicity of its eigenvalue λ as the *multiplicity* of λ.

We recall that for each $k \in [n]$, the k-th *Gershgorin's disk* of a matrix $\mathbf{M} = \{m_{ij}\}$ is defined as

$$D_k := \left\{ z \in \mathbb{C} : |z - m_{kk}| \leq \sum_{\substack{j=1 \\ j \neq k}}^{n} |m_{kj}| \right\}$$

and that each eigenvalue λ_i of $\mathbf{M}$, $i \in [n]$, satisfies the following result (Horn and Johnson, 2013):

Theorem 2 (Gershgorin's circle theorem)**.**

$$\lambda_i \in \bigcup_{k\in[n]} D_k.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ and $\lambda_1, \lambda_2 \ldots, \lambda_n$ denote the eigenvectors and eigenvalues, respectively, of a given matrix $\mathbf{M}$. In addition, define $\mathcal{S}_1 = \{0\}$ and, for each $k \in [2, n]$, denote by $S_k = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}$ and by $S_k^\perp$ the orthogonal complement of S_k. Then, if the eigenvalues of $\mathbf{M}$ satisfy

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

the following result holds (Horn and Johnson, 2013):

Theorem 3 (Courant–Fischer's theorem)**.**

$$\lambda_k = \min_{\substack{\boldsymbol{x}\in S_k^\perp \\ \boldsymbol{x}\neq \boldsymbol{0}}} \frac{\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{x}} = \min_{\substack{\boldsymbol{x}\in S_k^\perp \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} \quad \forall\, k \in [n]. \tag{1}$$

Observe that the maximum of $\min_{\substack{\mathbf{x}\in S \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{M}\mathbf{x}$ is achieved when $S = S_k^\perp$. This fact allows to rewrite (1) as

$$\lambda_k = \min_{\substack{T\subset\mathbb{R}^n \\ \dim(T)=n-k+1}} \max_{\substack{\mathbf{x}\in T \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{M}\mathbf{x} = \max_{\substack{\mathcal{S}\subset\mathbb{R}^n \\ \dim(\mathcal{S})=k}} \min_{\substack{\mathbf{x}\in S \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{M}\mathbf{x}.$$

The following results will prove useful in the next sections.

Proposition 1. *The matrix $\boldsymbol{M} = c\boldsymbol{1}_n\boldsymbol{1}_n^\top$, for some strictly positive real c and integer n, has eigenvalues*

- 0 *with algebraic multiplicity* $n-1$;
- cn *with algebraic multiplicity* 1.

Proof. As $\mathbf{M}$ has rank 1, by Theorem 1, its kernel has size $n-1$. Hence, $\mathbf{M}$ has a zero eigenvalue with multiplicity $n-1$. In addition, observe that

$$\mathbf{M}\cdot\mathbf{1}_n = c\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n) = cn\mathbf{1}_n. \tag{2}$$

Thus, $\mathbf{1}_n$ is an eigenvector of $\mathbf{M}$ and cn is its associated eigenvalue. □

Proposition 2. *Consider two symmetric and commuting matrices, $\boldsymbol{M}_1$ and $\boldsymbol{M}_2$, of order $n > 0$ and the block matrix*

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_1 & \boldsymbol{M}_2 \\ \boldsymbol{M}_2 & \boldsymbol{M}_1 \end{pmatrix} \tag{3}$$

of order $2n$. Let $\{\lambda_i\}$ and $\{\mu_i\}$ denote the eigenvalues of $\boldsymbol{M}_1$ and $\boldsymbol{M}_2$, respectively. In addition, denote by

$$\boldsymbol{M}' = \begin{pmatrix} \boldsymbol{M}_1 & \boldsymbol{M}_2 & \boldsymbol{M}_2 \\ \boldsymbol{M}_2 & \boldsymbol{M}_1 & \boldsymbol{M}_2 \\ \boldsymbol{M}_2 & \boldsymbol{M}_2 & \boldsymbol{M}_1 \end{pmatrix} \tag{4}$$

a block matrix of order $3n$. Then,

(i) $\boldsymbol{M}$ *has eigenvalues*

- $\lambda_i + \mu_i$ *with multiplicity* 1, *for all* $i \in [n]$;
- $\lambda_i - \mu_i$ *with multiplicity* 1, *for all* $i \in [n]$;

(ii) $\boldsymbol{M}'$ *has eigenvalues*

- $\lambda_i + 2\mu_i$ *with multiplicity* 1, *for all* $i \in [n]$;
- $\lambda_i - \mu_i$ *with multiplicity* 2, *for all* $i \in [n]$.

Proof. (i) Consider the following square matrix of order $2n$

$$\mathbf{U} = \frac{1}{\sqrt{2}}\begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{I}_n \end{pmatrix} \tag{5}$$

and observe that it is orthogonal, a $\mathbf{U}$ is real and its column vectors form an orthonormal system. Now, consider the matrix

$$\begin{aligned}
\mathbf{U}^{-1}\mathbf{M}\mathbf{U} &= \frac{1}{2}\begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{I}_n \end{pmatrix}\begin{pmatrix} \mathbf{M}_1 & \mathbf{M}_2 \\ \mathbf{M}_2 & \mathbf{M}_1 \end{pmatrix}\begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{I}_n \end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix} \mathbf{M}_1+\mathbf{M}_2 & \mathbf{M}_1+\mathbf{M}_2 \\ \mathbf{M}_1-\mathbf{M}_2 & \mathbf{M}_2-\mathbf{M}_1 \end{pmatrix}\begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{I}_n \end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 2(\mathbf{M}_1+\mathbf{M}_2) & \mathbf{0} \\ \mathbf{0} & 2(\mathbf{M}_2-\mathbf{M}_1) \end{pmatrix} = \begin{pmatrix} \mathbf{M}_1+\mathbf{M}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2-\mathbf{M}_1 \end{pmatrix}
\end{aligned}$$

and observe that $\mathbf{U}^{-1}\mathbf{M}\mathbf{U}$ and $\mathbf{M}$ are similar matrices, hence they have the same eigenvalues. In addition, observe that the set of eigenvalue of a block-diagonal matrix is the union of the eigenvalues of its blocks. As both $\mathbf{M}_1$ and $\mathbf{M}_2$ are, by hypothesis, symmetric and commuting, the statement follows.

(ii) To prove the statement, it is sufficient to show that there is a change of basis that transforms matrix $\mathbf{M}'$ into the form

$$\begin{pmatrix} \mathbf{M}_1 + 2\mathbf{M}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_1 - \mathbf{M}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{M}_1 - \mathbf{M}_2 \end{pmatrix}.$$

To this end, first consider the matrix

$$\mathbf{1}_3 \cdot \mathbf{1}_3^\top - \mathbf{I}_3 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \tag{6}$$

and observe that its eigenvalues are $\nu_1 = 2$ and $\nu_2 = -1$ with multiplicity 1 and 2, respectively. The unit-norm eigenvector associated with ν_1 is

$$\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right)^T$$

and the orthonormal eigenvectors associated with ν_2 are

$$\left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{3} \right)^\top \qquad \text{and} \qquad \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0 \right)^\top.$$

Then, the matrix

$$\mathbf{V} = \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{3} & 0 \end{pmatrix}$$

is orthogonal and diagonalizes $\mathbf{1}_3 \cdot \mathbf{1}_3^\top - \mathbf{I}_3$. Now, rewrite $\mathbf{M}'$ as $\mathbf{M}' = \mathbf{A} + \mathbf{B}$, where

$$\mathbf{A} = \begin{pmatrix} \mathbf{M}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{M}_1 \end{pmatrix} \qquad \text{and} \qquad \mathbf{B} = \begin{pmatrix} \mathbf{0} & \mathbf{M}_2 & \mathbf{M}_2 \\ \mathbf{M}_2 & \mathbf{0} & \mathbf{M}_2 \\ \mathbf{M}_2 & \mathbf{M}_2 & \mathbf{0} \end{pmatrix}$$

and observe that the following matrix

$$\hat{\mathbf{V}} = \begin{pmatrix} \frac{\sqrt{3}}{3}\mathbf{I}_n & \frac{\sqrt{6}}{6}\mathbf{I}_n & \frac{\sqrt{2}}{2}\mathbf{I}_n \\ \frac{\sqrt{3}}{3}\mathbf{I}_n & \frac{\sqrt{6}}{6}\mathbf{I}_n & -\frac{\sqrt{2}}{2}\mathbf{I}_n \\ \frac{\sqrt{3}}{3}\mathbf{I}_n & -\frac{\sqrt{6}}{3}\mathbf{I}_n & \mathbf{0} \end{pmatrix}$$

diagonalizes $\mathbf{B}$. As $\mathbf{U}^\top\mathbf{A}\mathbf{U} = \mathbf{A}$ for any orthogonal matrix $\mathbf{U}$ of order n, we conclude that

$$\hat{\mathbf{V}}^\top \mathbf{M}' \hat{\mathbf{V}} = \begin{pmatrix} \mathbf{M}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{M}_1 \end{pmatrix} + \begin{pmatrix} 2\mathbf{M}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{M}_2 \end{pmatrix}.$$

Similarly to the argument used in the proof of statement (i), as the set of eigenvalues of a block-diagonal matrix is the union of the eigenvalues of its blocks and both $\mathbf{M}_1$ and $\mathbf{M}_2$ are, by hypothesis, symmetric and commuting, the statement follows. $\square$

Trees, UBTs, and their matrix representations. We provide some further notation, definitions, and results for both general trees and UBTs that will prove useful in the reminder of the article. We start by denoting

- $\mathcal{U}_n$ as the set of the $(2n-5)!!$ possible UBTs with $n \geq 3$ leaves;
- Θ_n as the set of the $(2n-5)!!$ possible PLMs induced by the UBTs in $\mathcal{U}_n$;
- Δ_n as the set of the $(2n-5)!!$ possible PLDMs induced by the UBTs in $\mathcal{U}_n$;

A *cherry* of a UBT in $\mathcal{U}_n$ is a subgraph induced by two leaves and their adjacent (internal) vertex. Observe that, by definition of a UBT, any UBT $T \in \mathcal{U}_n$ has at least 2 cherries. The *diameter* of a tree T is the longest path (measured in terms of number of edges) between any pair of its leaves. A *caterpillar* $T \in \mathcal{U}_n$ is a UBT whose diameter is maximum, namely equal to $n-1$. For some $k \in \mathbb{N}$ and $n = 3 \cdot 2^k$, a *most-balanced UBT* in $\mathcal{U}_n$ is a UBT having a minimum diameter, namely equal to $2k+2$.

The following theorem from Catanzaro et al. (2025) characterizes Θ_n:

Theorem 4. *An integer matrix τ of order $n \geq 3$ encodes the PLM of a UBT $T \in \mathcal{U}_n$ if and only if it satisfies*

$$\begin{aligned}
\tau_{ij} &\in [2, n-1] && \forall \text{ distinct } i,j \in [n] && \text{(integrality conditions)}\\
\tau_{ji} &= \tau_{ij} && \forall \text{ distinct } i,j \in [n] && \text{(symmetry conditions)}\\
\tau_{ij} + \tau_{jk} - \tau_{ik} &\geq 2 && \forall \text{ distinct } i,j,k \in [n] && \text{(strong triangle inequalities)}\\
\sum_{\substack{j \in [n]\\ j \neq i}} 2^{-\tau_{ij}} &= \frac{1}{2} && \forall\, i \in [n] && \text{(Kraft's equalities)}
\end{aligned}$$

and, for all distinct $i,j,p,q \in [n]$, exactly one of the following

$$\begin{aligned}
&\tau_{ij} + \tau_{pq} + 2 \leq \tau_{ip} + \tau_{jq} = \tau_{iq} + \tau_{jp}\\
&\tau_{ip} + \tau_{jq} + 2 \leq \tau_{ij} + \tau_{pq} = \tau_{iq} + \tau_{jp} \qquad \text{(strong four-point conditions)}\\
&\tau_{iq} + \tau_{jp} + 2 \leq \tau_{ij} + \tau_{pq} = \tau_{ip} + \tau_{jq}.
\end{aligned}$$

We also recall the following result from Catanzaro et al. (2024):

Proposition 3. *Let $\mathbf{2}^{-\boldsymbol{\tau}}$ denote the PLDM induced by any UBT $T \in \mathcal{U}_n$. Then, there exist at least four distinct $i,j,k,l \in [n]$ such that $2^{-\tau_{ij}} = 2^{-\tau_{kl}} = \frac{1}{4}$. In addition, the following two conditions hold:*

$$\begin{aligned}
\tau_{ir} &= \tau_{jr} \quad \forall\, r \in [n] \setminus \{i,j\},\\
\tau_{kr} &= \tau_{lr} \quad \forall\, r \in [n] \setminus \{k,l\}.
\end{aligned}$$

Finally, we say that any two distinct rows i and j of a PLDM $\mathbf{2}^{-\boldsymbol{\tau}}$ induced by any UBT $T \in \mathcal{U}_n$ are *twins* if they satisfy $2^{-\tau_{ij}} = \frac{1}{4}$ and $\tau_{ir} = \tau_{jr}$ for all $r \in [n] \setminus \{i,j\}$.

3. The eigenspaces of PLDMs

In this section, we study the spectrum of the PLDMs induced by the UBTs in Θ_n. In particular, in Section 3.1 we study the relationships between some substructures of UBTs and the rational eigenvalues of the induced PLDMs. In Section 3.2, we characterize the spectra of both the caterpillar and the most-balanced UBT. Finally, in Subsection 3.3, we determine specific bounds for the eigenvalues of PLDMs induced by UBTs as well as their asymptotic behavior as n tends to infinity.

3.1. On the relationships between UBTs and the eigenvalues of the induced PLDMs

We study here the relationships between some substructures of UBTs and the rational eigenvalues of the induced PLDMs. We start by observing that the following proposition holds:

Proposition 4. Let $\mathbf{2}^{-\boldsymbol{\tau}}$ denote the PLDM induced by a UBT $T \in \mathcal{U}_n$. Then,

(i) For each entry $2^{-\tau_{ij}} = 1/4$, $i, j \in [n]$, $i < j$, the matrix $\mathbf{2}^{-\boldsymbol{\tau}}$ has an eigenvalue $-\frac{1}{4}$, with associated unit-norm eigenvalue $\mathbf{v} = \frac{\sqrt{2}}{2}(\mathbf{e}_i - \mathbf{e}_j)$.

(ii) The greatest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}}$ is $\frac{1}{2}$, and its associated unit-norm eigenvector is $\frac{\sqrt{n}}{n}\mathbf{1}_n$.

Proof.
(i). As T is a UBT, it has at least 2 cherries; hence, the matrix $\mathbf{2}^{-\boldsymbol{\tau}}$ has at least 4 entries equal to $1/4$. Without loss of generality, suppose that $2^{-\tau_{1,2}} = 1/4$. Then, we have

$$2^{-\tau}\left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, \ldots, 0\right]^{\top} = \left[-\frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{2}\left(2^{-\tau_{3,1}} - 2^{-\tau_{3,2}}\right), \ldots, \frac{\sqrt{2}}{2}\left(2^{-\tau_{n,1}} - 2^{-\tau_{n,2}}\right)\right]^{\top}. \tag{8}$$

As $2^{-\tau_{1,2}} = 1/4$, by Proposition 3 we have that $2^{-\tau_{k,1}} = 2^{-\tau_{k,2}}$, for all $k \in [3, n]$. Hence, (8) is equivalent to

$$2^{-\tau}\left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, \ldots, 0\right]^{\top} = -\frac{1}{4}\left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, \ldots, 0\right]^{\top}.$$

Thus, $\mathbf{2}^{-\boldsymbol{\tau}}$ has an eigenvalue $-\frac{1}{4}$, with an associated eigenvector $\mathbf{v}$.

(ii). Observe that each row of $\mathbf{2}^{-\boldsymbol{\tau}}$ sums to $\frac{1}{2}$ by Kraft's equalities, which implies that each row of $\mathbf{2}^{-\boldsymbol{\tau}} - \frac{1}{2}\mathbf{I}_n$ sums to 0. In turn, this implies that $\det(\mathbf{2}^{-\boldsymbol{\tau}} - \frac{1}{2}\mathbf{I}_n) = 0$, and thus $\frac{1}{2}$ is an eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}}$. Theorem 2 then ensures $\frac{1}{2}$ is the greatest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}}$. □

Proposition 4 establishes that each cherry is associated with the eigenvalue $-\frac{1}{4}$. We can generalize Proposition 4 as follows.

Given a UBT T, we refer to any of its cherries as a 0-*th order group of leaves* of T. Moreover, if the subgraph of T induced by two cherries is connected, we refer to it as a 1-*st order group of leaves* of T. Generally, we refer to the subgraph T' of T induced by two $(i-1)$-th order group of leaves as the i-*th order group of leaves* if T' is connected.

Theorem 5. *If $\mathbf{2}^{-\boldsymbol{\tau}}$ is the PLDM of a UBT T, then $\mathbf{2}^{-\boldsymbol{\tau}}$ has:*

- 1 *eigenvalue equal to* $\frac{2^d - 3}{2^{d+1}}$ *for each d-th order group of leaves;*
- 1 *eigenvalue equal to* $\frac{1}{2}$.

Proof. Assume a UBT with a d-th order group of leaves. Further assume that these leaves are the first 2^{d+1} leaves, arranged so that recursive bisection of this ordered set always produces an order subgroup of leaves. For example, dividing the first and second groups of $2^{d+1}/2$ leaves yields two separate groups of leaves of order $d-1$.

Let A be the PLDM of the UBT. The proposition is proven by observing that

$$\mathbf{v}_d = [\underbrace{1, 1, \dots, 1}_{2^{d+1}/2 \text{ entries}}, \overbrace{-1, -1, \dots, -1}^{2^{d+1}/2 \text{ entries}}, 0, 0 \dots, 0]^\top$$

is an eigenvector of A with eigenvalue $\frac{2^d-3}{2^{d+1}}$. Moreover, $\mathbf{v}_d$ is orthonormal to all eigenvectors of similar form that are associated with subgroups of leaves of smaller order, as defined by the recursive bisections of the leaves. □

3.2. Spectrum of most-balanced trees and caterpillars

PLDMs of most-balanced UBTs and caterpillars are characterized by specific spectra. We shall show that, despite the extremal and opposite structure of such two classes of trees, the spectra of their associated PLDMs underlie a common matrix construction.

Without loss of generality, let $\mathbf{2}^{-\boldsymbol{\tau}}$ be a PLDM of a UBT T with n leaves and $m \leq \frac{n}{2}$ cherries corresponding to the first $2m$ rows of $\mathbf{2}^{-\boldsymbol{\tau}}$, and let $\mathcal{M} = \{2r-1 : r \in [1, m]\}$ be the set of its first m odd rows. We first denote the *PLDM compression* of $\mathbf{2}^{-\boldsymbol{\tau}}$ as the matrix $\mathbf{W}^{-\boldsymbol{\tau}} = [w_{ij}]_{i,j \in [n-k]}$ as

$$\mathbf{W}^{-\boldsymbol{\tau}} = \mathbf{Q}\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U}, \tag{9}$$

where $\mathbf{Q}$ and $\mathbf{U}$ satisfy

- $\mathbf{Q}$ is an $(n-k) \times n$ matrix such that $w_{\frac{r+1}{2}\,r} = 1$ for $r \in \mathcal{M}$, $w_{m+i,2m+i} = 1$ for $i \in [1, n-2m]$, and $w_{ij} = 0$ otherwise.
- $\mathbf{U}$ is an $n \times (n-k)$ matrix such that $w_{r\,\frac{r+1}{2}} = w_{r+1,\,\frac{r+1}{2}} = 1$ for $r \in \mathcal{M}$, $w_{2m+i\,m+i} = 1$ for $i \in [1,\, n-2m]$, and $w_{ij} = 0$ otherwise,

or, equivalently,

- $(\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{\cdot r} = \mathbf{2}^{-\boldsymbol{\tau}}_{\cdot r} + \mathbf{2}^{-\boldsymbol{\tau}}_{\cdot r+1}$ for $r \in \mathcal{M}$, and $(\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{\cdot j} = \mathbf{2}^{-\boldsymbol{\tau}}_{\cdot j}$ for $j \in [2m+1, n]$.
- $(\mathbf{Q}\,\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{i\cdot} = (\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{2j-1,\cdot}$ for $j \in [1, m]$, and $(\mathbf{Q}\,\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{i\cdot} = (\mathbf{2}^{-\boldsymbol{\tau}}\mathbf{U})_{i\cdot}$ for $i \in [2m+1,\, n]$.

Observe that the hypothesis of having the first $2m$ rows of $\mathbf{2}^{-\boldsymbol{\tau}}$ associated to the m cherries of T is not restrictive, as PLDMs that can be obtained from one another through symmetric permutations of rows and columns are associated with the same UBT up to a permutation of the labels of the leaves. Hence, we hereafter denote the PLDM projection of a PLDM $\mathbf{2}^{-\boldsymbol{\tau}}$ as the PLDM projection $\mathbf{W}^{-\boldsymbol{\tau}}$ associated with the PLDM obtained from $\mathbf{2}^{-\boldsymbol{\tau}}$ through a suitable sequence of symmetric permutations.

We show that $\mathbf{W}^{-\boldsymbol{\tau}}$ is almost similar to $\mathbf{2}^{-\boldsymbol{\tau}}$.

Proposition 5. *Let $\mathbf{2}^{-\boldsymbol{\tau}}$ be a PLDM of a UBT T with n leaves and m cherries, and let $\mathbf{W}^{-\boldsymbol{\tau}}$ its PLDM projection. Then, $\mathbf{W}^{-\boldsymbol{\tau}}$ has the same eigenvalues of $\mathbf{2}^{-\boldsymbol{\tau}}$, except for those equal to $-\frac{1}{4}$ associated to the m cherries.*

Proof. To be completed. This result is an immediate consequence of the structure of the eigenvectors associated with the eigenvalue $-\frac{1}{4}$. □

We clarify Proposition 5 through the following example.

Example 1. *Consider the following PLDM associated with a UBT having 8 leaves and 3 cherries:*

$$\boldsymbol{2}^{-\boldsymbol{\tau}} = \begin{pmatrix} 0 & 2^{-2} & 2^{-5} & 2^{-5} & 2^{-6} & 2^{-6} & 2^{-3} & 2^{-5} \\ 2^{-2} & 0 & 2^{-5} & 2^{-5} & 2^{-6} & 2^{-6} & 2^{-3} & 2^{-5} \\ 2^{-5} & 2^{-5} & 0 & 2^{-2} & 2^{-5} & 2^{-5} & 2^{-4} & 2^{-4} \\ 2^{-5} & 2^{-5} & 2^{-2} & 0 & 2^{-5} & 2^{-5} & 2^{-4} & 2^{-4} \\ 2^{-6} & 2^{-6} & 2^{-5} & 2^{-5} & 0 & 2^{-2} & 2^{-5} & 2^{-3} \\ 2^{-6} & 2^{-6} & 2^{-5} & 2^{-5} & 2^{-2} & 0 & 2^{-5} & 2^{-3} \\ 2^{-3} & 2^{-3} & 2^{-4} & 2^{-4} & 2^{-5} & 2^{-5} & 0 & 2^{-4} \\ 2^{-5} & 2^{-5} & 2^{-4} & 2^{-4} & 2^{-3} & 2^{-3} & 2^{-4} & 0 \end{pmatrix}$$

By defining

$$\boldsymbol{Q} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{U} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Matrix $\boldsymbol{W}^{-\boldsymbol{\tau}}$ in the statement of Proposition 5 is given by

$$\boldsymbol{W}^{-\boldsymbol{\tau}} = \boldsymbol{Q}\boldsymbol{2}^{-\boldsymbol{\tau}}\boldsymbol{U} = \begin{pmatrix} 2^{-2} & 2^{-4} & 2^{-5} & 2^{-3} & 2^{-5} \\ 2^{-4} & 2^{-2} & 2^{-4} & 2^{-4} & 2^{-4} \\ 2^{-5} & 2^{-4} & 2^{-2} & 2^{-5} & 2^{-3} \\ 2^{-2} & 2^{-3} & 2^{-4} & 0 & 2^{-4} \\ 2^{-4} & 2^{-3} & 2^{-2} & 2^{-4} & 0 \end{pmatrix}$$

Matrix $\mathbf{W}^{-\boldsymbol{\tau}}$ has several properties. It is easy to see that $w_{ij} > 0$ for all $i, j \in [1, n-m]$, $i \neq j$, and

$$w_{ii} = \begin{cases} \frac{1}{4} & \text{if } i \leq m, \\ 0 & \text{if } m+1 \leq i \leq n. \end{cases}$$

In addition, $\sum_{j=1}^{n-m} w_{ij} = \frac{1}{2}$, and thus its greatest eigenvalue is $\frac{1}{2}$. The following properties also hold.

Proposition 6. *Let $\boldsymbol{W}^{-\boldsymbol{\tau}}$ be the PLDM compression of $\boldsymbol{2}^{-\boldsymbol{\tau}}$, and let $\boldsymbol{Q}$ and $\boldsymbol{U}$ be its presentation. Then, $\boldsymbol{Q}\boldsymbol{U} = \boldsymbol{I}_{n-m}$. In addition, $\boldsymbol{W}^{-\boldsymbol{\tau}}$ has the block structure:*

$$\boldsymbol{W}^{-\boldsymbol{\tau}} = \begin{pmatrix} \boldsymbol{W}_{11} & \boldsymbol{W}_{21} \\ \boldsymbol{W}_{21} & \boldsymbol{W}_{22} \end{pmatrix}$$

Blocks $\boldsymbol{W}_{11}$ and $\boldsymbol{W}_{22}$ are symmetric, and $\boldsymbol{W}_{21} = 2\,\boldsymbol{W}_{12}^{\top}$. In particular, all the diagonal entries of $\boldsymbol{W}_{11}$ are equal to $\frac{1}{4}$ and the non-diagonal entry (i, j) $\boldsymbol{W}_{11}$ is equal to $2^{-\tau_{2i-1,2j-1}+1}$ holds.

Consider the $(n-m) \times (n-m)$ matrix $V = U^T U$, V is an diagonal matrix such that

$$v_{ii} = \begin{cases} 2 & \text{if } i \in [1, m], \\ 1 & \text{otherwise} \end{cases}.$$

Consequently,

- Matrix $V^{\frac{1}{2}} W V^{-\frac{1}{2}}$ is symmetric and similar to W, so have the same eigenvalues.

- Matrix VWV^{-1} is similar to W, so have the same eigenvalues, but not symmetric.
- Matrix WV^{-1} and VW are symmetric but not similar.

The symmetry properties of the above matrices derive by the block structure of W and are exploited in the following propositions.

Proposition 7. *Let $B(k)$ denote the $2^{-\tau}$ matrix associated with the most balanced UBT with $n = 3 \cdot 2^k$ taxa and $m = 3 \cdot 2^{k-1}$ cherries, and so 2^{k-1} eigenvalues equal to $-\frac{1}{4}$. Let a $2^{-\tau} = B(k)$, then the matrix W satisfies*

$$W = QB(k)U = V^{-1}B(k-1) + \frac{1}{4}I_{3\cdot 2^{k-1}}.$$

Proof. The statement is an immediate consequence of observing that W coincides with its sole block W_{11} and that $V = 2I_{3\cdot 2^{k-1}}$. □

The following result is an immediate consequence of Proposition 7.

Theorem 6. *Given an integer $k > 0$, if $\boldsymbol{2}^{-\boldsymbol{\tau}}$ is the PLDM of a most-balanced UBT T with $n = 3 \cdot 2^k$ taxa, then $\boldsymbol{2}^{-\boldsymbol{\tau}}$ has eigenvalues*

- *$\frac{2^d-3}{2^{d+1}}$ with multiplicity $3 \cdot 2^{k-d}$ for each $d \in [1, k]$,*
- *$\frac{2^{k+1}-3}{2^{k+2}}$ with multiplicity 2,*
- *$\frac{1}{2}$ with multiplicity 1,*

that is, $k+2$ different eigenvalues such that $\lambda_s = -\frac{1}{4} + 3\sum_{r=1}^{s} \frac{1}{2^{r+2}}$, for $s \in [1, k+1]$, and $\lambda_{k+2} = \frac{1}{2}$.

Proof. By recursively applying Proposition 7, we obtain that $B(k)$ has:

- $3 \cdot 2^{k-1}$ eigenvalues
$$\lambda = -\frac{1}{4},$$
- $3 \cdot 2^{k-2}$ eigenvalues
$$\lambda = \frac{1}{4} + \frac{1}{2}\left(-\frac{1}{4}\right) = \frac{1}{8},$$
- $3 \cdot 2^{k-3}$ eigenvalues
$$\lambda = \frac{1}{4} + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{2}\left(-\frac{1}{4}\right)\right) = \frac{5}{16},$$

and, in general, $3 \cdot 2^{k-d}$ eigenvalues

$$\lambda = \sum_{i=2}^{d} \frac{1}{2^i} - \frac{1}{2^{d+1}} = \frac{2^d - 3}{2^{d+1}}, \qquad d \in [1, k].$$

Finally, since $B(0)$ has two eigenvalues equal to $-\frac{1}{4}$ and one equal to $\frac{1}{2}$, it follows that $B(k)$ has two eigenvalues equal to $\frac{2^{k+1}-3}{2^{k+2}}$, and one eigenvalue equal to $\frac{1}{2}$. □

An alternative proof of Theorem 6 can be found in Appendix A.

Now, let $2^{-\tau}$ be the the matrix associated with the standard caterpillar UBT with n taxa and $m = 2$ cherries, where the first cherry is made by taxa 1 and 2, and the second cherry is made by taxa $n-1$ and n. Modify the structure of matrices Q and U in Proposition **??** accordingly and observe that V is then the diagonal matrix such that

$$v_{ii} = \begin{cases} 2 & \text{if } i = 1 \text{ or } i = n-2, \\ 1 & \text{otherwise} \end{cases}.$$

Proposition 8. *Given a matrix $2^{-\tau}$ be with the standard caterpillar UBT with n taxa. Then,*

$$W = Q2^{-\tau}U = \frac{1}{4}(SV - I_{n-2}) \tag{10}$$

where S is $(n-2) \times (n-2)$ dyadic matrix such that $s_{ij} = 2^{-|i-j|}$.

Proof. Since $QU = I_{n-2}$, it can be directly verified that the following conditions hold:

$$2^{-\tau} = \frac{1}{4}(USU^T - I_{n-2}) \quad \Leftrightarrow \quad W = Q2^{-\tau}U = \frac{1}{4}(QUSU^TU - QI_{n-2}U) = \frac{1}{4}(SV - I_{n-2}).$$

$\square$

Condition 10 can then be used to determine the eigenvalues of W for caterpillars.

Toward a characterization of the spectrum of PLDMs of caterpillars, we first recall the following results on *Kac-Murdock-Szegö matrices*, i.e., those matrices with the form $\left[\rho^{-|i-j|}\right]_{i,j\in[n]}$ for $\rho \geq 0$, introduced by Kac et al. (1953) and later generalized by Fikioris (2018) (cfr. Theorem 6.5 and Theorem 2.1 in (Fikioris, 2018)).

Proposition 9. *Let $\boldsymbol{S} = \left[\frac{1}{2^{|i-j|}}\right]_{i,j\in[n]}$ and $\gamma_1 \leq \gamma_2 \leq \dots \gamma_n$ be its eigenvalues. Then,*

$$\gamma_i = \frac{3}{5 - 4\cos(\chi_i)}, i \in [n], \tag{11}$$

where χ_i is given by the unique root

$$x : \begin{cases} 2\cos\frac{(n+1)x}{2} = \cos\frac{(n-1)x}{2} & \textit{if } i \bmod 2 = 0 \\ 2\sin\frac{(n+1)x}{2} = \sin\frac{(n-1)x}{2} & \textit{otherwise} \end{cases}, \qquad \textit{for } x \in \left[\frac{i\pi}{n}, \frac{(i+1)\pi}{n+1}\right] \tag{12}$$

Proposition 10. *Let $\boldsymbol{S} = \left[\frac{1}{2^{|i-j|}}\right]_{i,j\in[n]}$. Then, $\boldsymbol{S}$ is invertible and*

$$\boldsymbol{S}^{-1} = \frac{4}{3}\begin{pmatrix} 1 & -\frac{1}{2} & 0 & 0 & 0 & \dots & 0 \\ -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} & 0 & 0 & \dots & 0 \\ 0 & -\frac{1}{2} & \frac{5}{4} & \ddots & 0 & \dots & 0 \\ \vdots & 0 & \ddots & \ddots & \ddots & & \vdots \\ & \vdots & \ddots & \ddots & \frac{5}{4} & -\frac{1}{2} & 0 \\ & & & & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} \\ 0 & \dots & \dots & & 0 & -\frac{1}{2} & 1 \end{pmatrix}$$

We are now ready to state the following result, by building upon some ideas used in the proof of Proposition 9 (Kac et al., 1953; Fikioris, 2018) and (da Fonseca and Petronilho, 2001; Tan, 2019).

Theorem 7. *Given a PLDM $\mathbf{2}^{-\boldsymbol{\tau}}$ of order $n \geq 4$ associated with a caterpillar T, the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ of $\mathbf{2}^{-\boldsymbol{\tau}}$ are given by*

$$\lambda_i = \frac{1}{4}(\mu_i - 1), \quad i \in [n] \tag{13}$$

such that $\mu_1 = \mu_2 = 0$ and

$$\mu_i = \frac{3}{5 - 4\cos\alpha_i}, \quad i \in [3, n] \tag{14}$$

where $\{\alpha_i : i \in [3, n]\}$ are the solutions of

$$\tan\left(\frac{(n-3)\alpha}{2}\right) = \frac{1}{3}\cot\left(\frac{\alpha}{2}\right) \quad \vee \quad \cot\left(\frac{(n-3)\alpha}{2}\right) = -3\tan\left(\frac{\alpha}{2}\right), \tag{15}$$

for $\alpha \in [0, \pi)$.

Proof. Without loss of generality, suppose that $\mathbf{2}^{-\boldsymbol{\tau}}$ has the form

$$\mathbf{2}^{-\boldsymbol{\tau}} = \begin{pmatrix} 0 & 2^{-2} & 2^{-3} & \ldots & 2^{2-n} & 2^{1-n} & 2^{1-n} \\ 2^{-2} & 0 & 2^{-3} & \ldots & 2^{2-n} & 2^{1-n} & 2^{1-n} \\ 2^{-3} & 2^{-3} & 0 & \ldots & 2^{3-n} & 2^{2-n} & 2^{2-n} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 2^{2-n} & 2^{2-n} & 2^{3-n} & \ldots & 0 & 2^{-3} & 2^{-3} \\ 2^{1-n} & 2^{1-n} & 2^{2-n} & \ldots & 2^{-3} & 0 & 2^{-2} \\ 2^{1-n} & 2^{1-n} & 2^{2-n} & \ldots & 2^{-3} & 2^{-2} & 0 \end{pmatrix}. \tag{16}$$

Let

$$\mathbf{U} = \begin{pmatrix} \mathbf{I}_{1\cdot} \\ \mathbf{I}_{n-2} \\ \mathbf{I}_{n\cdot} \end{pmatrix}.$$

Observe that $\mathbf{U} \in \mathbb{Z}^{n \times (n-2)}$ Let also $\mathbf{S} = [2^{-|i-j|}]_{i,j \in [n-2]}$. Then, observe that

$$\mathbf{U}\mathbf{S}\mathbf{U}^\top = \begin{pmatrix} 1 & \mathbf{S}_{1\cdot} & \mathbf{2}^{-n+1} \\ \mathbf{S}_{\cdot 1} & \mathbf{S} & \mathbf{S}_{\cdot n} \\ \mathbf{2}^{-n+1} & \mathbf{S}_{n\cdot} & 1 \end{pmatrix}$$

which entails

$$\mathbf{2}^{-\boldsymbol{\tau}} = \frac{1}{4}\left(\mathbf{U}\mathbf{S}\mathbf{U}^\top - \mathbf{I}_n\right). \tag{17}$$

Hence,

$$\lambda_i = \frac{1}{4}\left(\mu_i - 1\right), \quad i \in [n], \tag{18}$$

We have that $(\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{\cdot 1} = (\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{\cdot 2}$ and $(\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{\cdot n-1} = (\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{\cdot n}$. Hence $[1\ -1\, 0 \cdots 0]^\top$ and $[0 \cdots 0\ 1\ {-1}]^\top$ are two eigenvectors of $\mathbf{U}\mathbf{S}\mathbf{U}^\top$ associated with its eigenvalues $\mu_1 = 0$ and $\mu_2 = 0$. In turn, μ_1 and μ_2 correspond to the two smallest eigenvalues $\lambda_1 = -\frac{1}{4}$ and $\lambda_2 = -\frac{1}{4}$ of $\mathbf{2}^{-\boldsymbol{\tau}}$, respectively.

All the remaining eigenvectors of $\mathbf{U}\mathbf{S}\mathbf{U}^\top$ not associated with $\mu_1 = \mu_2 = 0$ must be orthogonal to $[1\ \ -1\ \ 0 \cdots 0]^\top$ and $[0 \cdots 0\ \ 1\ \ -1]^\top$ and so they are of the form $v = [v_2\ v_2\ v_3 \cdots v_{n-2}\ v_{n-1}\ v_{n-1}]^\top$, i.e., both their first and the two entries must be identical.

Since $(\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{1\cdot} = (\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{2\cdot}$ and $(\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{n-1\cdot n-1} = (\mathbf{U}\mathbf{S}\mathbf{U}^\top)_{n\cdot}$ also hold, the following condition must hold for any eigenvalue μ distinct from $\mu_1 = \mu_2 = 0$:

$$\mathbf{U}\mathbf{S}\mathbf{U}^\top \mathbf{v} = \mu \mathbf{v} \quad \Leftrightarrow \quad (\mathbf{S}_{\cdot 1} \mid \mathbf{S} \mid \mathbf{S}_{\cdot n-2}) \begin{pmatrix} v_2 \\ v_2 \\ v_3 \\ \vdots \\ v_{n-2} \\ v_{n-1} \\ v_{n-1} \end{pmatrix} = \mu \begin{pmatrix} v_2 \\ v_3 \\ \vdots \\ v_{n-2} \\ v_{n-1} \end{pmatrix} \quad \Leftrightarrow \quad \mathbf{V}\mathbf{S}\mathbf{w} = \mu \mathbf{w} \quad (19)$$

where the diagonal matrix V and the vector w are defined as follows:

$$V = \begin{pmatrix} 2 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 2 \end{pmatrix}, \qquad w = \begin{pmatrix} v_2 \\ v_3 \\ \vdots \\ v_{n-2} \\ v_{n-1} \end{pmatrix}.$$

As $\mathbf{V}$ is diagonal, $\mathbf{S}$ is symmetric, and both $\mathbf{V}$ and $\mathbf{S}$ are invertible, (19) is in turn equivalent to

$$\mathbf{V}^{-1}\mathbf{S}^{-1}\mathbf{w} = \frac{1}{\mu}\mathbf{w}. \tag{20}$$

In particular, we have

$$\begin{aligned}
\mathbf{V}^{-1}\mathbf{S}^{-1} &= \frac{4}{3} \begin{pmatrix} \frac{1}{2} & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & -\frac{1}{2} & & & & \\ -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} & & & \\ & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} & & \\ & & \ddots & \ddots & \ddots & \\ & & & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} \\ & & & & -\frac{1}{2} & 1 \end{pmatrix} \\
&= \frac{4}{3} \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & & & & \\ -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} & & & \\ & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} & & \\ & & \ddots & \ddots & \ddots & \\ & & & -\frac{1}{2} & \frac{5}{4} & -\frac{1}{2} \\ & & & & -\frac{1}{4} & \frac{1}{2} \end{pmatrix},
\end{aligned}$$

where $\mathbf{S}^{-1}$ has the form given by Proposition 10. Hence, (20) is equivalent to the following discrete Sturm-Liouville problem with boundary conditions:

$$\begin{cases} \dfrac{1}{\mu} w_1 = \dfrac{4}{3}\left(\dfrac{1}{2} w_1 - \dfrac{1}{4} w_2\right) & \text{(21a)} \\ \dfrac{1}{\mu} w_k = \dfrac{4}{3}\left(-\dfrac{1}{2} w_{k-1} + \dfrac{5}{4} w_k - \dfrac{1}{2} w_{k+1}\right) \quad \forall\, k \in [1, n-3] & \text{(21b)} \\ \dfrac{1}{\mu} w_{n-2} = \dfrac{4}{3}\left(\dfrac{1}{2} w_{n-2} - \dfrac{1}{4} w_{n-3}\right) & \text{(21c)} \end{cases}$$

Solving (21) is equivalent to solving the recurrence

$$\left(\frac{5\mu-3}{2\mu}\right) w_k = w_{k-1} + w_{k+1} \tag{22}$$

subject to the boundary conditions (21a) and (21c).

Observe that the eigenvalues $\frac{1}{\mu_i}$ of $\mathbf{V}^{-1}\mathbf{S}^{-1}$ satisfy

$$\frac{1}{3} \le \frac{1}{\mu_i} \le 3 \;\Leftrightarrow\; \frac{1}{3} \le \mu_i \le 3 \tag{23}$$

for each $i \in [n-2]$ by Theorem 2. Thus,

$$\left|\frac{5\mu_i-3}{2\mu_i}\right| \le 2, \quad \forall\, i \in [3,n],$$

which entails that there is at least a value of α satisfying the following identity:

$$2\cos\alpha = \frac{5\mu_i-3}{2\mu_i}, \quad 0 \le \alpha < 2\pi, \quad \forall\, i \in [3,n]. \tag{24}$$

Now, consider the characteristic equation associated with (22):

$$x^2 - (2\cos\alpha)x + 1 = 0, \tag{25}$$

implying that

$$x_k = c'\sin(k\alpha+\phi) \tag{26}$$

for some $c', \phi \in \mathbb{R}$. Now, let

$$\begin{aligned} \beta &= \alpha+\phi, \\ \beta' &= (n-2)\alpha+\phi, \end{aligned} \tag{27}$$

Now, consider conditions (21a) and (21c):

$$\begin{aligned} &\begin{cases} \frac{1}{\mu}w_1 = \frac{4}{3}\left(\frac{1}{2}w_1 - \frac{1}{4}w_2\right) \\ \frac{1}{\mu}w_{n-2} = \frac{4}{3}\left(\frac{1}{2}w_{n-2} - \frac{1}{4}w_{n-3}\right) \end{cases} \\ \Leftrightarrow &\begin{cases} \frac{1}{\mu}\sin(\alpha+\phi) = \frac{4}{3}\left(\frac{1}{2}\sin(\alpha+\phi) - \frac{1}{4}\sin(2\alpha+\phi)\right) \\ \frac{1}{\mu}\sin((n-2)\alpha+\phi) = \frac{4}{3}\left(\frac{1}{2}\sin((n-2)\alpha+\phi) - \frac{1}{4}\sin((n-3)\alpha+\phi)\right) \end{cases} \\ \Leftrightarrow &\begin{cases} \sin(2\alpha+\phi) = \left(2-\frac{3}{\mu}\right)\sin(\alpha+\phi) \\ \sin((n-3)\alpha+\phi) = \left(2-\frac{3}{\mu}\right)\sin((n-2)\alpha+\phi) \end{cases} \end{aligned} \tag{28}$$

As (24) is equivalent to

$$\mu = \frac{3}{5-4\cos\alpha}$$

and by definitions (27), system (28) can be written as

$$\begin{cases} \sin(\beta+\alpha) = (4\cos\alpha-3)\sin\beta \\ \sin(\beta'-\alpha) = (4\cos\alpha-3)\sin\beta' \end{cases} \tag{29}$$

Suppose that $\beta = k\pi \ \forall \ k \in \mathbb{Z}$. Hence, $\sin\beta = 0$ and $\cos\beta = \pm 1$, and thus (29) is equivalent to

$$\begin{cases} \sin\beta\cos\alpha + \sin\alpha\cos\beta = (4\cos\alpha - 3)\sin\beta \\ \sin\beta'\cos\alpha - \sin\alpha\cos\beta' = (4\cos\alpha - 3)\sin\beta' \end{cases}$$
$$\Leftrightarrow \begin{cases} \sin\alpha = 0 \\ \sin\beta'\cos\alpha = (4\cos\alpha - 3)\sin\beta' \end{cases}$$
$$\Leftrightarrow \begin{cases} \alpha = \pi \\ \pm\sin\beta' = (4\cos\alpha - 3)\sin\beta' \end{cases} \tag{30}$$

Hence, $\phi = (k-1)\pi$. Then, if $\beta' = k'\pi \ \forall \ k \in \mathbb{Z}$ we have

$$\begin{aligned} &\sin\beta' = 0 \\ &\Leftrightarrow \ \beta' = k'\pi \ \forall \ k' \in \mathbb{Z} \\ &\Leftrightarrow \ (n-2)\pi + (k-1)\pi = k'\pi \ \forall \ k, k' \in \mathbb{Z} \\ &\Leftrightarrow \ n = 3 + k' - k \ \forall \ k, k' \in \mathbb{Z}, \end{aligned}$$

implying that $\sin\beta' = 0$ is satisfied by all $n \geq 3$. Conversely, if $\beta' \neq k'\pi$, we obtain that (30) is equivalent to

$$\begin{cases} \alpha = \pi \\ 4\cos\alpha - 3 = \pm 1 \end{cases} \tag{31}$$

Hence, in both cases, system (29) is satisfied by $\alpha = \pi$ and $\phi = (k-1)\pi$. By definition, if $4\cos\alpha - 3 = 1$ then $\mu = 3$, corresponding to the eigenvalue $\frac{1}{2}$ of $\mathbf{2}^{-\boldsymbol{\tau}}$. Otherwise, if $4\cos\alpha - 3 = -1$, then $\mu = 1$, corresponding to the eigenvalue 0 of $\mathbf{2}^{-\boldsymbol{\tau}}$. Case $\beta \neq k\pi, \beta' = k'\pi, \forall \ k, k' \in \mathbb{Z}$ yields a similar result.

Suppose now that $\beta = \frac{\pi}{2} + k\pi, \forall \ k \in \mathbb{Z}$. In this case $\sin\beta = \pm 1$ and $\cos\beta = 0$, and thus (29) is equivalent to

$$\begin{cases} \sin\beta\cos\alpha + \sin\alpha\cos\beta = c\sin\beta \\ \sin\beta'\cos\alpha - \sin\alpha\cos\beta' = c\sin\beta' \end{cases}$$
$$\Leftrightarrow \begin{cases} \cos\alpha = 4\cos\alpha - 3 \\ c\sin\beta' - \sin\alpha\cos\beta' = c\sin\beta' \end{cases}$$
$$\Leftrightarrow \begin{cases} \cos\alpha = 4\cos\alpha - 3 \\ \sin\alpha\cos\beta' = 0 \end{cases}$$

Then, $\alpha = 2k\pi \ \forall \ k \in \mathbb{Z}$ and thus the system is satisfied for each $\beta' \in \mathbb{R}$.

We are left to consider the two following cases: $\beta, \beta' \neq k\pi \ \forall \ k \in \mathbb{Z}$ and $\beta, \beta' \neq \frac{\pi}{2} + k\pi \ \forall \ k \in \mathbb{Z}$ (case (b)).

Case (a). System (29) is equivalent to

$$\begin{cases} \tan\beta = -\frac{\sin\alpha}{3(1-\cos\alpha)} \\ \tan\beta' = \frac{\sin\alpha}{3(1-\cos\alpha)} \end{cases} \tag{32}$$

which entails

$$\tan(\beta' - \beta) = \frac{\tan\beta' - \tan\beta}{1 + \tan\beta'\tan\beta} = \frac{2\tan\beta'}{1 + \tan^2\beta'} = \tan 2\beta'. \tag{33}$$

Then, since $\beta' - \beta = (n-3)\alpha$, (33) implies

$$\tan 2\beta' = \tan((n-3)\alpha) \;\Leftrightarrow\; 2\beta' = (n-3)\alpha,$$

Hence, by recalling that

$$\tan\frac{\alpha}{2} = \frac{\sin\alpha}{1+\cos\alpha} = \frac{1-\cos\alpha}{\sin\alpha}, \tag{34}$$

we obtain

$$\tan\frac{((n-3)\alpha)}{2} = \tan\beta' = \frac{1}{3}\frac{\sin\alpha}{1-\cos\alpha} = \frac{1}{3\tan\frac{\alpha}{2}}. \tag{35}$$

Case (b). System (29) is equivalent to

$$\begin{cases} \cot\beta = -\frac{3(1-\cos\alpha)}{\sin\alpha)} \\ \cot\beta' = \frac{3(1-\cos\alpha)}{\sin\alpha} \end{cases}, \tag{36}$$

which entails

$$\cot(\beta' - \beta) = \frac{\cot\beta'\cot\beta + 1}{\cot\beta - \cot\beta'} = -\frac{\cot^2\beta' + 1}{2\cot\beta'} = -\cot 2\beta'. \tag{37}$$

Then, since $\beta' - \beta = (n-3)\alpha$, (37) implies

$$-\cot 2\beta' = \cot((n-3)\alpha) \;\Leftrightarrow\; \pi - 2\beta' = (n-3)\alpha,$$

By recalling (34), we obtain

$$\cot\left(\frac{(n-3)\alpha}{2}\right) = \cot(\pi - \beta') = -3\frac{\cos(\pi-\alpha)-1}{\sin(\pi-\alpha)} = 3\frac{\cos\alpha - 1}{\sin\alpha} = -3\tan\left(\frac{\alpha}{2}\right), \tag{38}$$

which, together with condition (35), is equivalent to (15). In addition, by symmetry, the distinct solutions of (35) and (38) are in the interval $[0,\pi)$. Function $\tan\left(\frac{(n-3)\alpha}{2}\right)$ has period $\frac{2\pi}{n-3}$, and given $\gamma = \frac{(2k+1)\pi}{n-3}$, it is monotone increasing in $(\gamma, \gamma + \frac{\pi}{n-3}$, arbitrarily small as $\alpha \to \gamma^-$, and arbitrarily large as $\alpha \to \gamma^+$ for each $k \in \mathbb{Z}$. On the other hand, function $\frac{1}{3}\cot\left(\frac{\alpha}{2}\right)$ is equal to $\tan\left(\frac{(n-3)\alpha}{2}\right)$ when $\alpha = 0$, it is monotonic decreasing in $[0,\pi)$ and arbitrarily small as $\alpha \to \pi^-$. Hence, (35) has exactly $\lceil\frac{n-3}{2}\rceil$ distinct zeros. A similar argument applied to $\cot\left(\frac{(n-3)\alpha}{2}\right)$ and $-3\tan(\frac{\alpha}{2})$ yields exactly $\lfloor\frac{n-3}{2}\rfloor$ distinct zeros for equation (38). □

Unfortunately, it is very difficult to generalize the previous arguments to determine closed formulas for eigenvalues of a generic PLDM. However, we can provide a recursive characterization that may be useful computationally. Toward this end, we recall the following two results reported by Horn and Johnson (2013) (cfr. Corollary 4.3.9, p. 241 and Section 0.8, p. 25, respectively).

Proposition 11. *Given a symmetric real matrix M of order $n \geq 2$ and $v \in \mathbb{R}^n$, let $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$ be the eigenvalues of M. Then, the eigenvalues $\mu_1 \leq \mu_2 \leq \ldots \mu_n$ of $M + vv^\top$ satisfy:*

$$\lambda_i \leq \mu_i \leq \lambda_{i+1}.$$

Proposition 12 (Schur's complement)**.** *Given an integer $n \geq 2$, let $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ be the* *block matrix*

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}.$$

Then, if $\boldsymbol{A}$ *is invertible,*

$$\det(\boldsymbol{M}) = \det(\boldsymbol{A}) \det(\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}).$$

In addition, if $\boldsymbol{D}$ *is invertible,*

$$\det(\boldsymbol{M}) = \det(\boldsymbol{D}) \det(\boldsymbol{A} - \boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C}).$$

We are now ready to state the recursive characterization of the eigenvalues of $\mathbf{2}^{-\boldsymbol{\tau}}$.

Proposition 13. *Given an integer* $n \geq 4$*, let* $\boldsymbol{2}^{-\boldsymbol{\tau}}$ *be a PLDM of order* n *having the form*

$$\boldsymbol{2}^{-\boldsymbol{\tau}} = \begin{pmatrix} \boldsymbol{M} & \frac{1}{2}\boldsymbol{b} & \frac{1}{2}\boldsymbol{b} \\ \frac{1}{2}\boldsymbol{b}^\top & 0 & -\frac{1}{4} \\ \frac{1}{2}\boldsymbol{b}^\top & -\frac{1}{4} & 0 \end{pmatrix} \tag{39}$$

where $\boldsymbol{M} \in \mathbb{R}^{(n-2)\times(n-2)}$ *and* $\boldsymbol{b} \in \mathbb{R}^{n-2}$*, and* $\boldsymbol{2}^{-\boldsymbol{\tau}'}$ *be the PLDM obtained through a contraction of* $\boldsymbol{2}^{-\boldsymbol{\tau}}$ *with respect to the equivalent rows* $n-1$ *and* n*. Let also* $\mu_1 \leq \mu_2 \leq \ldots \leq \mu_{n-2}$ *be the eigenvalues of* $\boldsymbol{M}$*, and*

$$\boldsymbol{U} \in \mathbb{R}^{(n-2)\times(n-2)} \; : \; \boldsymbol{U}^\top \boldsymbol{M} \boldsymbol{U} = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_{n-2}), \qquad \boldsymbol{c} = \frac{\sqrt{2}}{2}\boldsymbol{U}^\top \boldsymbol{b},$$

Then, the eigenvalues of $\boldsymbol{2}^{-\boldsymbol{\tau}}$ *are*

- $-\frac{1}{4}$ *with multiplicity 1;*
- *the zeros of the following equation:*

$$f(x) = \left(\prod_{i=1}^{n-2} (\mu_i - x)\right)\left(-\frac{1}{4} - x - \sum_{i=1}^{n-2} \frac{c_i^2}{\mu_i - x}\right), \tag{40}$$

- *each eigenvalue of* $\boldsymbol{2}^{-\boldsymbol{\tau}'}$ *having an associated eigenvector* $\boldsymbol{v}$ *such that* $0 = \boldsymbol{v}^\top \boldsymbol{b} = c_i$*.*

Proof. First, observe that any PLDM can be reduced to the form (39) through symmetric permutations of rows and columns. By hypothesis, we have:

$$\mathbf{2}^{-\boldsymbol{\tau}'} = \begin{pmatrix} \mathbf{M} & \mathbf{b} \\ \mathbf{b}^\top & 0 \end{pmatrix},$$

Now, consider matrix

$$\mathbf{V} = \begin{pmatrix} \mathbf{I}_{n-2} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

and let

$$\mathbf{T} = \mathbf{V}^\top 2^{-\boldsymbol{\tau}}\mathbf{V}. \tag{41}$$

Hence, to determine the eigenvalues of $\mathbf{2}^{-\boldsymbol{\tau}}$ it is sufficient to consider matrix (41). Observe that

$$\mathbf{T} = \begin{pmatrix} \mathbf{I}_{n-2} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} \mathbf{M} & \frac{1}{2}\mathbf{b} & \frac{1}{2}\mathbf{b} \\ \frac{1}{2}\mathbf{b}^\top & 0 & -\frac{1}{4} \\ \frac{1}{2}\mathbf{b}^\top & -\frac{1}{4} & 0 \end{pmatrix} \begin{pmatrix} \mathbf{I}_{n-2} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$
$$= \begin{pmatrix} \mathbf{M} & \frac{1}{2}\mathbf{b} & \frac{1}{2}\mathbf{b} \\ \frac{\sqrt{2}}{2}\mathbf{b}^\top & -\frac{\sqrt{2}}{8} & -\frac{\sqrt{2}}{8} \\ 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{2}}{8} \end{pmatrix} \begin{pmatrix} \mathbf{I}_{n-2} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} \mathbf{M} & \frac{\sqrt{2}}{2}\mathbf{b} & 0 \\ \frac{\sqrt{2}}{2}\mathbf{b}^\top & -\frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}, \tag{42}$$

Thus, the eigenvalues of $\mathbf{2}^{-\boldsymbol{\tau}}$ are $-\frac{1}{4}$ and those of the principal submatrix

$$\mathbf{S} = \begin{pmatrix} \mathbf{M} & \frac{\sqrt{2}}{2}\mathbf{b} \\ \frac{\sqrt{2}}{2}\mathbf{b}^\top & -\frac{1}{4} \end{pmatrix} \tag{43}$$

Let

$$\mathbf{Q} = \begin{pmatrix} \mathbf{U} & 0 \\ 0 & 1 \end{pmatrix}$$

Then, the eigenvalues of (43) are given by the solution of

$$\det(\mathbf{S} - \lambda \mathbf{I}_{n-1}) = 0 \quad \Leftrightarrow \quad \det(\mathbf{Q}^\top(\mathbf{S} - \lambda \mathbf{I}_{n-1})\mathbf{Q}) = 0$$
$$\Leftrightarrow \det \begin{pmatrix} \mathbf{U}^\top(\mathbf{M} - \lambda \mathbf{I}_{n-2})\mathbf{U} & \frac{\sqrt{2}}{2}\mathbf{U}^\top \mathbf{b} \\ \frac{\sqrt{2}}{2}\mathbf{b}^\top \mathbf{U} & -\frac{1}{4} - \lambda \end{pmatrix} = 0 \Leftrightarrow \det \begin{pmatrix} \mathrm{diag}(\mu_i - \lambda) & \mathbf{c} \\ \mathbf{c}^\top & -\frac{1}{4} - \lambda \end{pmatrix} = 0,$$

which, by applying Proposition 12, is equivalent to

$$\left(-\frac{1}{4} - \lambda - \sum_{i=1}^{n-2} \frac{c_i^2}{\mu_i - \lambda} \right) \prod_{i=1}^{n-2} (\mu_i - \lambda) = 0,$$

thereby proving the statement. □

3.3. Extremal properties of PLDMs

Theorems 6 and 7 state that $-\frac{1}{4}$ is the smallest eigenvalue of the PLDMs associated with most-balanced UBTs and caterpillars, respectively. It is natural to ask whether we can generalize this observation to all UBTs. Toward this end, we first recall the following result (cfr. Horn and Johnson (2013), Corollary 4.3.3, page 240), which states a property on interlacing eigenvalues of symmetric matrices when perturbed with another matrix satisfying some strong assumptions.

Proposition 14. *Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be two real symmetric matrices of order n such that $\boldsymbol{B}$ has exactly 1 positive and 1 negative eigenvalue. Let also $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$ and $\mu_1 \leq \mu_2 \leq \ldots \leq \mu_n$ be the eigenvalues of $\boldsymbol{A}$ and $\boldsymbol{A} + \boldsymbol{B}$, respectively. Then,*

$$\mu_i \leq \lambda_{i+1}, \quad \forall\, i \in [1, n-1], \tag{44}$$

with equality for $i \in [1, n-1]$ if and only if $\boldsymbol{B}$ is singular and there is a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{A}\boldsymbol{x} = \lambda_{i+1}\boldsymbol{x}$, $\boldsymbol{B}\boldsymbol{x} = 0$, and $(\boldsymbol{A} + \boldsymbol{B})\boldsymbol{x} = \mu_i \boldsymbol{x}$.

We also report Poincaré separation theorem (cfr. Horn and Johnson (2013), Corollary 4.3.37, page 248), that states a property on interlacing eigenvalues of two symmetric matrices when one can be obtained from the other through an orthogonal transformation.

Theorem 8 (Poincaré separation theorem)**.** *Given positive integers m and n with $m < n$, let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{m \times m}$ be symmetric, and let U be an orthogonal matrix such that*

$B = U^\top AU$. *Let also* $\lambda_1 \leq \lambda_2, \leq \ldots \leq \lambda_n$ *and* $\mu_1 \leq \mu_2 \leq \ldots \leq \mu_n$ *be the eigenvalues of* $\boldsymbol{A}$ *and* $\boldsymbol{B}$*, respectively. Then,*

$$\lambda_i \leq \mu_i \leq \lambda_{i+n-m} \quad \forall\ i \in [m].$$

Before proceeding, we also need to recall to definition of *contraction*, introduced in (Catanzaro et al., 2020b) and later used in (Catanzaro et al., 2024) polyhedral results for the convex hull of PLDMs for small values of n. Given a matrix $\mathbf{M} = [m_{ij}]_{i,j\in[n]}$, a contraction on M with respect to rows p and q, with $p < q$, yields a matrix M' obtained by first subtracting 1 from the non-diagonal entries of the q-th row and column of M and then zeroing out its p-th row and column, i.e.,

$$\mathbf{M}' = \begin{cases} 0 & \text{if } p = i \text{ or } q = i, \\ m_{pq} - 1 & \text{if } p \neq q \text{ and } ((p = j \text{ and } q \neq i) \text{ or } (q = j \text{ and } p \neq i)), \\ m_{pq} & \text{otherwise,} \end{cases} \tag{45}$$

In case M is a PLDM, contractions have a precise topological interpretation. Suppose that $M = 2^{-\boldsymbol{\tau}}$, and $2^{-\boldsymbol{\tau}'}$ is obtained from $2^{-\boldsymbol{\tau}}$ through a contraction with respect to the two equivalent rows i and j, with $i < j$ (i.e., two rows satisfying $2^{-\tau_{ij}} = \frac{1}{4}$ and (**??**); see Section 2). Then, $2^{-\boldsymbol{\tau}'}$ corresponds to a UBT T' obtained from the UBT T associated with $2^{-\boldsymbol{\tau}}$ by collapsing leaves i and j into their common internal node.

We can now prove that $-\frac{1}{4}$ is the smallest eigenvalues of all PLDMs.

Proposition 15. *Given a PLDM* $\boldsymbol{2}^{-\boldsymbol{\tau}}$ *of order* $n \geq 0$ *associated with a UBT* T*,* $-\frac{1}{4}$ *is the smallest eigenvalue of* $\boldsymbol{2}^{-\boldsymbol{\tau}}$*.*

Proof. First, observe that there exists a most-balanced UBT T' with $3 \cdot 2^{\lceil \frac{n}{3} \rceil}$ leaves and associated PLDM $\mathbf{2}^{-\boldsymbol{\tau}'}$ such that $\mathbf{2}^{-\boldsymbol{\tau}}$ can be obtained from $\mathbf{2}^{-\boldsymbol{\tau}'}$ through subsequent contractions.

By Proposition 6, there are i and j such that $2^{-\tau'_{ij}} = \frac{1}{4}$. We perform a contraction on $\mathbf{2}^{-\boldsymbol{\tau}'}$ with respect to i and j to obtain a matrix $\mathbf{2}^{-\boldsymbol{\tau}''}$ associated with a UBT T'', having the form

$$\mathbf{2}^{-\boldsymbol{\tau}''} = \mathbf{U}^\top(\mathbf{2}^{-\boldsymbol{\tau}'} + \mathbf{E})\mathbf{U}, \tag{46}$$

where $\mathbf{E}$ is a matrix having the j-th row (and column) equal to the j-th row (and column) of $\mathbf{2}^{-\boldsymbol{\tau}'}$,

$$\mathbf{E} = \mathbf{e}_j\mathbf{v}^\top + \mathbf{v}\mathbf{e}_j^\top, \qquad \mathbf{v} = \mathbf{2}^{-\boldsymbol{\tau}'}\mathbf{e}_j,$$

and $\mathbf{U}$ is the matrix in $\mathbb{Z}^{n\times(n-1)}$ having $\mathbf{e}_k$, $k \in [1, n] \setminus \{i\}$ as columns, i.e.,

$$\mathbf{U} = \begin{pmatrix}\mathbf{e}_1\ \mathbf{e}_2\ \ldots\ \mathbf{e}_{i-1}\ \mathbf{e}_{i+1}\ \ldots\ \mathbf{e}_n\end{pmatrix}$$

Observe that the rank of $\mathbf{E}$ is equal to 2, as the image of $\mathbf{E}$ is contained in the span of $\mathbf{e}_j$ and $\mathbf{v}$, which are linearly independent because $\mathbf{e}_j^\top \cdot \mathbf{v} = 0$. Hence, $\mathbf{E}$ has eigenvalue 0 with multiplicity greater than or equal to $\dim\ker(\mathbf{E}) = n - 2$. A full-rank representation of matrix $\mathbf{E}$ in the basis $\{\mathbf{e}_j, \mathbf{v}\}$ is given by a matrix $\mathbf{M}$ satisfying

$$\mathbf{E}[\mathbf{e}_j \quad \mathbf{v}] = [\mathbf{e}_j \quad \mathbf{v}]\mathbf{M},$$

i.e., the columns of $\mathbf{M}$ are the coordinates of $\mathbf{e}_j$ and $\mathbf{v}$ in the basis $\{\mathbf{e}_j, \mathbf{v}\}$. Observe that

$$\mathbf{E}\mathbf{e}_j = 0\mathbf{e}_j + 1\mathbf{v}, \quad \mathbf{E}\mathbf{v} = ||\mathbf{v}||^2\mathbf{e}_j + 0\mathbf{v},$$

and thus

$$\mathbf{M} = \begin{pmatrix} 0 & ||\mathbf{v}||^2 \\ 1 & 0 \end{pmatrix}$$

As eigenvalues are preserved by change of basis, the (non-zero) eigenvalues of $\mathbf{E}$ are given by

$$\det\begin{pmatrix} -\mu & ||\mathbf{v}||^2 \\ 1 & -\mu \end{pmatrix} = 0 \;\Leftrightarrow\; \mu^2 - ||\mathbf{v}||^2 = 0 \;\Leftrightarrow\; \mu = \pm||\mathbf{v}||$$

and so $\mathbf{E}$ has exactly 1 positive eigenvalue equal to $||\mathbf{v}||$, and 1 negative eigenvalue equal to $-||\mathbf{v}||$.

As $\mathbf{2}^{-\boldsymbol{\tau}'}$ is the PLDM of a most-balanced UBT, there are $k, l \in [n]$ such that $2^{-\tau'_{k,l}} = \frac{1}{4}$ and i, j, k, l are all distinct. Hence, by an argument similar to the proof of Proposition 4, $\mathbf{2}^{-\boldsymbol{\tau}'} + E$ has eigenvalue $\lambda = -\frac{1}{4}$ having multiplicity at least 1 and associated eigenvector $\mathbf{x}$ whose k-th and l-th entries are equal to c and $-c$ for some $c \in \mathbb{R}$, respectively, and 0 elsewhere. In addition, $\mathbf{E}$ is singular with $\mathbf{E}\mathbf{x} = \mathbf{0}_n$ as $\mathbf{e}_s^\top \mathbf{E}\mathbf{x}$ is equal to 0 for each $s \in [n] \setminus \{j, k\}$ by Proposition 3 and $(\mathbf{2}^{-\boldsymbol{\tau}'} + \mathbf{E})\mathbf{x} = \lambda\mathbf{x}$ by construction, and thus the smallest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}'} + \mathbf{E}$ is equal to the second smallest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}'}$, i.e., $\lambda = -\frac{1}{4}$, by Proposition 14. Then, as $\mathbf{U}$ is orthogonal by construction, the smallest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}''}$ is greater or equal to the smallest eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}'} + \mathbf{E}$ by Theorem 8, and this relation must hold to equality by Proposition 4.

We can iterate the same argument until obtaining T. Observing that the tree obtained at each iteration is a UBT by construction and the smallest eigenvalue of its associated PLDM is $-\frac{1}{4}$ with multiplicity greater than or equal to 2 proves the statement. ☐

As a direct application of Proposition 15, we show the converse of statement (a) of Proposition 4.

Proposition 16. *Let $2^{-\boldsymbol{\tau}}$ be a rational matrix of order $n \geq 3$ satisfying Kraft's equalities, strong triangle inequalities and having exactly $2d \geq 4$ entries equal to 2. Then, the multiplicity of the eigenvalue $-\frac{1}{4}$ of matrix $2^{-\boldsymbol{\tau}}$ is equal to d. In particular, each vector in the orthonormal basis of the eigenspace associated with $-\frac{1}{4}$ has the form $\frac{\sqrt{2}}{2}(\boldsymbol{e}_i - \boldsymbol{e}_j)$, where i and j are equivalent rows of $2^{-\boldsymbol{\tau}}$.*

Proof. (sketch) First, by Proposition 3, there are exactly d pairs of equivalent rows. Now, let

$$\mathbf{M} = 4 \cdot \mathbf{2}^{-\boldsymbol{\tau}} + \mathbf{I}_n.$$

Then, to prove the statement, it is sufficient to show that

$$\dim(\ker(\mathbf{M})) = d \tag{47}$$

Without loss of generality, suppose that the first $2d$ rows of matrix $\mathbf{2}^{-\boldsymbol{\tau}}$ are pairwise equivalent, where specifically rows $2i-1$ and $2i$ are equivalent for each $i \in [d]$. Let $\mathbf{U}$ be an orthogonal matrix in $\mathbb{R}^{n \times (n-d)}$ of the form

$$\mathbf{U} = \begin{pmatrix} 1 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & 1 & \dots & 0 & 0 \\ \vdots & & & & \ddots & & \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

and let

$$\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{U}^\top.$$

☐

Proposition 15 ensures that the smallest eigenvalue of the PLDM $2^{-\boldsymbol{\tau}}$ of any UBT is equal to $-\frac{1}{4}$. Propositions 15 and 16 ensure that the eigenvalue $-\frac{1}{4}$ corresponds to equivalent rows of $2^{-\boldsymbol{\tau}}$ or, equivalently, to cherries of T. Hereafter, we show that, in the case of PLDMs associated with caterpillars, $-\frac{1}{4}$ is the only eigenvalue less than or equal to $-\frac{1}{6}$.

Proposition 17. *Let $\mathbf{2}^{-\boldsymbol{\tau}}$ be the PLDM associated with a caterpillar with $n \geq 3$ leaves, and let $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$ be its eigenvalues. Then,*

$$-\frac{1}{6} < \lambda_i < \frac{1}{2} \quad \forall\, i \in [3, n-1].$$

Proof. By Theorem 7, $\lambda_i = \frac{1}{4}\left(\frac{3}{5-4\cos\alpha_i} - 1\right)$ for each $i \in [3, n-1]$, which in turn implies the statement as $-1 < \cos\alpha_i < 1$. $\square$

Interestingly, this result can be generalized to the PLDM of any UBT through interlacing arguments.

Theorem 9. *Let $\mathbf{2}^{-\boldsymbol{\tau}}$ be the PLDM associated with a UBT T with $n \geq 5$ leaves and d cherries, and let $\nu_1 \leq \nu_2 \leq \ldots \leq \nu_n$ be its eigenvalues. Then,*

$$-\frac{1}{6} < \nu_i < \frac{1}{2}, \quad \forall\, i \in [d+1, n-1].$$

Proof. Apply zero or more contractions on $\mathbf{2}^{-\boldsymbol{\tau}}$ by iterating the construction in the proof of Proposition 15 until obtaining a tree T' with an associated PLDM $\mathbf{2}^{-\boldsymbol{\tau}'}$ such that there exists a contraction on $\mathbf{2}^{-\boldsymbol{\tau}'}$ that yields a PLDM $\mathbf{2}^{-\boldsymbol{\tau}''}$ associated with a caterpillar T'' (it is easy to see that this is always possible through 0 or more contractions). Let μ_i be the i-th eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}''}$. By Proposition 17,

$$\mu_1 = \mu_2 = -\frac{1}{4}, \quad \mu_3 > -\frac{1}{6}.$$

Now, recall that $\mathbf{2}^{-\boldsymbol{\tau}''}$ has the form (46). Let λ_i be the i-th eigenvalue of $\mathbf{2}^{-\boldsymbol{\tau}'} + \mathbf{E}$. Then, by the same argument used in Proposition 15, λ_1, λ_2, λ_3 are equal to the second, the third and the fourth eigenvalues of $\mathbf{2}^{-\boldsymbol{\tau}'}$ i.e.,

$$\lambda_1 = \lambda_2 = \lambda_3 = -\frac{1}{4}$$

as T' has exactly three cherries. Suppose, by contradiction, that $\lambda_4 \leq -\frac{1}{6}$. Then, by Theorem 8, we have

$$-\frac{1}{6} < \mu_3 \leq \lambda_4 \leq -\frac{1}{6},$$

a contradiction. $\square$

In the case of caterpillars, we can determine the extremal properties of the eigenvalues $-\frac{1}{6}$ and $\frac{1}{4}$. To this end, we recall the following notation and results. Let i be the imaginary unit and, given a subset $S \subseteq \mathbb{R}^n$, denote the *Lebesgue measure* of S by $\mathcal{L}(S)$. Given a Toeplitz matrix $\mathbf{T}_n = (t_{jk})_{j,k\in[n]}$, a bounded periodic function $f : [0, 2\pi] \to \mathbb{R}$ is called a *symbol* of $\mathbf{T}_n$ if its Fourier coefficients

$$c_l = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-il\theta}d\theta, \quad l \in \mathbb{Z},$$

are such that $t_{jk} = c_{|j-k|}$. For a given symmetric real square matrix $\mathbf{M}$ of order n having eigenvalues $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$, we denote by $N_{\mathbf{M}}(x)$ the sum of the multiplicities of the eigenvalues of $\mathbf{M}$ that are less than or equal to x, i.e.,

$$N_{\mathbf{M}}(x) = |\{j \in [n] : \lambda_j \leq x\}|.$$

Then, we have the following formulation of Szegő first limit theorem (Gray, 2001), given by Böttcher et al. (2010).

Theorem 10 (Szegő first limit theorem). *Let $f : [0, 2\pi] \to \mathbb{R}$ be a symbol of the Toeplitz matrix $\boldsymbol{T}_n = (t_{jk})_{j,k\in[n]}$, and let $\lambda_1^{(n)} \leq \cdots \leq \lambda_n^{(n)}$ be the eigenvalues of $\boldsymbol{T}_n$. Then, for every interval (α, β) such that*

$$\mathcal{L}\big(\{\theta \in [0, 2\pi] : f(\theta) = \alpha\}\big) = 0, \qquad \mathcal{L}\big(\{\theta \in [0, 2\pi] : f(\theta) = \beta\}\big) = 0,$$

the following holds

$$\frac{1}{n}\big|\{j : \lambda_j^{(n)} \in (\alpha, \beta)\}\big| = \frac{1}{2\pi}\mathcal{L}(\{\theta \in [0, 2\pi] : f(\theta) \in (\alpha, \beta)\}) + o(1).$$

We also have the following related result (Horn and Johnson, 2013).

Proposition 18 (Bai-Silverstein rank inequality). *Given two symmetric square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of order n,*

$$\sup_{x\in\mathbb{R}} (N_{\boldsymbol{A}}(x) - N_{\boldsymbol{B}}(x)) \leq \mathit{rank}(\boldsymbol{A} - \boldsymbol{B})$$

We can now show that $-\frac{1}{6}$ and $\frac{1}{2}$ are both accumulation points and provide their extremal ratio.

Proposition 19. *Given a PLDM $2^{-\boldsymbol{\tau}}$ of a caterpillar T with n leaves, the ratio between the sums of the multiplicities of the eigenvalues in the open right and left intervals with arbitrarily small length centered on $\frac{1}{2}$ and $-\frac{1}{6}$, respectively, approaches $\frac{1}{9}$ as $n \to \infty$.*

Proof. Without loss of generality, suppose that $2^{-\boldsymbol{\tau}}$ has the form (16). Then,

$$\mathbf{2}^{-\boldsymbol{\tau}} = \mathbf{T}_n + \mathbf{E},$$

where $\mathbf{T}_n$ is the Toeplitz matrix

$$\mathbf{T}_n = \begin{cases} 2^{-|i-j|-2} & \forall\, i \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

and $\mathbf{E}$ is a symmetric real matrix whose only nonzero rows and columns are the first and the last, and thus

$$\operatorname{rank}(\mathbf{E}) \leq 4.$$

Then, by Proposition 18,

$$\sup_{x\in\mathbb{R}}(N_{\mathbf{2}^{-\boldsymbol{\tau}}}(x) - N_{\mathbf{T}_n}(x)) \leq 4$$

Hence, for any $0 < \epsilon < \frac{1}{4}$ and sufficiently large n, we have

$$\frac{N_{\mathbf{2}^{-\boldsymbol{\tau}}}(\frac{1}{2}) - N_{\mathbf{2}^{-\boldsymbol{\tau}}}(\frac{1}{2} - \epsilon)}{N_{\mathbf{T}_n}(\frac{1}{2}) - N_{\mathbf{T}_n}(\frac{1}{2} - \epsilon)} = \frac{N_{\mathbf{2}^{-\boldsymbol{\tau}}}(-\frac{1}{6} + \epsilon) - N_{\mathbf{2}^{-\boldsymbol{\tau}}}(-\frac{1}{6})}{N_{\mathbf{T}_n}(-\frac{1}{6} + \epsilon) - N_{\mathbf{T}_n}(-\frac{1}{6})} + \mathcal{O}(1),$$

which entails

$$\begin{aligned}&\lim_{\epsilon\to 0^+}\lim_{n\to\infty}\frac{N_{\mathbf{2}^{-}\boldsymbol{\tau}}(\frac{1}{2})-N_{\mathbf{2}^{-}\boldsymbol{\tau}}(\frac{1}{2}-\epsilon)}{N_{\mathbf{2}^{-}\boldsymbol{\tau}}(-\frac{1}{6}+\epsilon)-N_{\mathbf{2}^{-}\boldsymbol{\tau}}(-\frac{1}{6})}\\ &=\lim_{\epsilon\to 0^+}\lim_{n\to\infty}\frac{N_{\mathbf{T}_n}(\frac{1}{2})-N_{\mathbf{T}_n}(\frac{1}{2}-\epsilon)}{N_{\mathbf{T}_n}(-\frac{1}{6}+\epsilon)-N_{\mathbf{T}_n}(-\frac{1}{6})}.\end{aligned}\tag{48}$$

To prove the claim, it is sufficient to prove that Let $c_k=\frac{1}{2^k}$ for $k\in\mathbb{Z}$, and consider the Fourier series

$$f(\theta)=\sum_{k\in\mathbb{Z}}c_k e^{-ik\theta}=\sum_{k=1}^{\infty}\frac{1}{2^k}\cos(k\theta)\tag{49}$$

By standard properties of geometric series, (49) is equivalent to

$$f(\theta)=\frac{1}{2}\frac{\frac{1}{2}\left(\cos\theta-\frac{1}{2}\right)}{1-\cos(\theta)+\frac{1}{4}}=\frac{\cos\theta-\frac{1}{2}}{5-4\cos(\theta)}\tag{50}$$

Observe that (50) is monotonic non-increasing for $\theta\in[0,\pi]$, and

$$f(0)=\frac{1}{2},\quad f(\pi)=-\frac{1}{6}.$$

Consider again (48). By Theorem 10, the following equality holds:

$$\begin{aligned}&\lim_{\epsilon\to 0^+}\lim_{n\to\infty}\frac{N_{\mathbf{T}_n}(\frac{1}{2})-N_{\mathbf{T}_n}(\frac{1}{2}-\epsilon)}{N_{\mathbf{T}_n}(-\frac{1}{6}+\epsilon)-N_{\mathbf{T}_n}(-\frac{1}{6})}\\ &=\lim_{\epsilon\to 0^+}\lim_{n\to\infty}\frac{\mathcal{L}(\{\theta\in[0,2\pi]:f(\theta)\in(\frac{1}{2}-\epsilon,\frac{1}{2}]\})+o(1)}{\mathcal{L}(\{\theta\in[0,2\pi]:f(\theta)\in(-\frac{1}{6},-\frac{1}{6}+\epsilon]\})+o(1)}\end{aligned}\tag{51}$$

Let $\theta^-=\{\theta\in[0,2\pi]:f(\theta)\in(\frac{1}{2}-\epsilon,\frac{1}{2}]\}$. Observe that the Taylor expansion of $\cos\theta$ around $\theta=0$ is given by

$$\cos\theta=1-\frac{\theta^2}{2}+\mathcal{O}(\theta^4),$$

and then, as it can be easily verified,

$$f(\theta)=\frac{1}{2}-\frac{3}{2}\theta^2+\mathcal{O}(\theta^4),$$

which entails

$$f(\theta)\geq\frac{1}{2}-\epsilon\;\Rightarrow\;\frac{3}{2}\theta^2\geq\epsilon+\mathcal{O}(\theta^4)\;\Rightarrow\;\mathcal{L}(\theta^-)=\sqrt{\frac{2}{3}\epsilon}+o(1)$$

Similarly, let $\theta^+=\{\theta\in[0,2\pi]:f(\theta)\in(-\frac{1}{6},-\frac{1}{6}+\epsilon]\}$ and let $\phi=\theta-\pi$. Then,

$$\cos(\theta)=\cos(\pi+\phi)=-\cos(\phi)$$

The Taylor expansion of $\cos\theta$ around $\theta=\pi$ is equal to the Taylor expansion of $-\cos(\phi)$ around 0, i.e.,

$$\cos(\theta)=-1+\frac{(\theta-\pi)^2}{2}+\mathcal{O}(\theta^4),$$

and then, as it can be easily verified,

$$f(\theta) = -\frac{1}{6} + \frac{1}{54}(\theta-\pi)^2 + \mathcal{O}((\theta-\pi)^4),$$

which entails

$$f(\theta) \le -\frac{1}{6} + \epsilon \;\Rightarrow\; \frac{1}{54}(\theta-\pi)^2 \le \epsilon + \mathcal{O}((\theta-\pi)^4) \;\Rightarrow\; \theta^+ = 3\sqrt{6\epsilon} + o(1)$$

Hence, (51) is equal to

$$\lim_{\epsilon\to 0^+}\lim_{n\to\infty} \frac{\mathcal{L}(\theta^-)+o(1)}{\mathcal{L}(\theta^+)+o(1)} = \lim_{\epsilon\to 0^+} \frac{\sqrt{\frac{2}{3}\epsilon}+o(1)}{3\sqrt{6\epsilon}+o(1)} = \frac{1}{9},$$

proving the statement. □

3.4. Relation between the diameter of a UBT and the spectrum of its associated PLDM

Consider matrix $2^{-\tau}$ associated to the UBT T. Observe that $A = 2\cdot 2^{-\tau}$ is a stochastic matrix with spectrum $\sigma(A)$, which obviously include $\lambda_{\max} = 1$.

The Dobrushin coefficient, denoted as $\delta(A)$, provides an upper bound on the magnitudes of all eigenvalues except the dominant one, that is

$$\max\{|\lambda| : \lambda \in \sigma(A) \setminus \{1\}\} \le \delta(A) \tag{52}$$

The Dobrushin coefficient is defined in terms of the entries of the transition matrix A as:

$$\delta(A) = 1 - \min_{i<j} \sum_{k=1}^{n} \min\{p_{ik}, p_{jk}\} = 2\left(\frac{1}{2} - \min_{i<j} \sum_{k=1}^{n} \min\{2^{-\tau_{ik}}, 2^{-\tau_{jk}}\}\right) \tag{53}$$

Conditions (52) and (53) imply that all the eigenvalues $\lambda \neq \frac{1}{2}$, and in particular the second largest eigenvalue $\lambda_\star$, of $2^{-\tau}$ satisfy the inequality

$$|\lambda| \le \frac{1}{2}\delta(A) = \frac{1}{2} - \min_{i<j} \sum_{k=1}^{n} \min\{2^{-\tau_{ik}}, 2^{-\tau_{jk}}\} = \frac{1}{2} - \sum_{k=1}^{n} \min\{2^{-\tau_{i_0 k}}, 2^{-\tau_{j_0 k}}\} \tag{54}$$

where i_0 and j_0 are extreme nodes of the diameter of the UBT T.

Consider each internal node r of a diameter of T of length m with extreme nodes i_0 and i_0. Denote by $S(r)$ the subtree S_r rooted in a node r of the diameter. The following conditions hold:

- either i_0 is farther than j_0 from all the nodes that belong to S_r or vice versa.
- $\sum_{k\in S_r} 2^{\tau_{kr}} = \frac{1}{2}$

The above conditions imply that:

$$\begin{aligned}
&\sum_{k=1}^{n} \min\{2^{-\tau_{i_0 k}}, 2^{-\tau_{j_0 k}}\} \\
&\quad = \begin{cases} 2\sum_{r=\lfloor\frac{m}{2}\rfloor+2}^{m} 2^{-r} = 2^{-\lfloor\frac{m}{2}\rfloor} - 2^{-(m-1)} & m \text{ is odd} \\ 2\sum_{r=\frac{m}{2}+2}^{m} 2^{-r} + 2^{-(\frac{m}{2}+1)} = 2^{-\frac{m}{2}} + 2^{-(\frac{m}{2}+1)} - 2^{-(m-1)} & m \text{ is even} \end{cases}
\end{aligned}$$

4. Final remarks and open problems

In this article, we showed that the spectrum of PLDMs of UBTs reveals significant information on their topology. We proved that most-balanced trees and caterpillars are completely determined by their spectrum. In the intermediate cases, specific substructures are determined by rational eigenvalues. Given a UBT $T \in \mathbb{U}_n$, a natural question is if there exists some information on T that, complemented with the spectrum of its associated PLDM, allows one to uniquely encode the topology of T with space complexity that is linear in n, i.e., equivalent to listing all edges explicitly.

Question 1. *Given an integer $n \geq 3$ and a tree $T \in \mathbb{U}_n$ with associated PLDM $\boldsymbol{2}^{-\boldsymbol{\tau}}$, what is the space complexity of the minimum amount of information on the topology of T that, together with the eigenvalues of $\boldsymbol{2}^{-\boldsymbol{\tau}}$, is sufficient to encode T?*

Theorem 4 encodes the set of PLMs through algebraic conditions on their entries for $n \geq 3$, and other characterizations exist when $3 \leq n \leq 11$ (Catanzaro et al., 2024). On the other hand, Theorems 6 and 7 give a strong characterization of the spectrum of PLDMs associated with most-balanced UBTs and caterpillars. We ask whether such conditions, complemented with Kraft's equalities for topological consistency, are sufficient to encode the PLMs of these specific types of trees among all integer matrices of order n.

Conjecture 1. *If $\boldsymbol{\tau}$ is an integer matrix of order $n \geq 3$ satisfying Kraft's equalities and the eigenvalues of $\boldsymbol{2}^{-\boldsymbol{\tau}}$ are*

a) as the ones in the statement of Theorem 6, then $\boldsymbol{2}^{-\boldsymbol{\tau}}$ is the PLDM of a most-balanced UBT;

b) as the ones in the statement of Theorem 7, then $\boldsymbol{2}^{-\boldsymbol{\tau}}$ is the PLDM of a caterpillar.

For sufficiently small values of n, the number of pairs of trees having cospectral distance matrices is a very small fraction of the total number of distinct topologies (Matsen and Evans, 2012), suggesting that the knowledge of the spectrum, combined with other constant-space local information on the topology, may be sufficient to encode small trees. Such a result may prove useful in applications of combinatorial optimization (Chung, 1995; Mohar and Poljak, 1993). In particular, consider the case of the *Balanced Minimum Evolution Problem* (BMEP) that, given a symmetric real-valued square matrix $\mathbf{D} = [d_{ij}]_{i,j \in [n]}$, requires finding the UBT T whose associated PLDM $\mathbf{2}^{-\boldsymbol{\tau}}$ minimizes:

$$L(\mathbf{2}^{-\boldsymbol{\tau}}) = \sum_{i,j \in [n], i \neq j} d_{ij} 2^{-\tau_{ij}}.$$

The following question arises naturally.

Question 2. *Given an instance $\mathcal{I}$ of the BMEP with matrix $\boldsymbol{D}$ having optimal solution $\boldsymbol{2}^{-\boldsymbol{\tau}^{\star}}$, what is the class of perturbations on matrix $\boldsymbol{D}$ that yields BMEP instances having $\boldsymbol{2}^{-\boldsymbol{\tau}^{\star}}$ as their optimal solution?*

Appendix A. Alternative proof of Theorem 6

Proof. We prove the statement by induction. If $k = 0$, then

$$\mathbf{2}^{-\boldsymbol{\tau}} = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{pmatrix},$$

and the statement is easily verified. Hence, suppose that the statement is true for some integer $k > 0$. Observe that, in this case, $\mathbf{2}^{-\boldsymbol{\tau}}$ has the form given by (4), i.e.,

$$\mathbf{2}^{-\boldsymbol{\tau}} = \begin{pmatrix} \mathbf{A} & \mathbf{B} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{B} & \mathbf{A} \end{pmatrix}$$

for some matrices $\mathbf{A}$ and $\mathbf{B}$ of order $\frac{n}{3}$, as T can be partitioned into three distinct rooted binary trees with the same topology. Moreover, $\mathbf{B} = 2^{-(2k+2)}\mathbf{1}\mathbf{1}^\top$, as the distance between one of the leaves in one of the three rooted binary trees and one of the leaves of any of the two other rooted binary trees is equal to $2k + 2$.
Let $\alpha_1 \leq \alpha_2 \leq \ldots \leq \alpha_{n/3}$ and $\beta_1 \leq \beta_2 \leq \ldots \leq \beta_{n/3}$ be the eigenvalues of $\mathbf{A}$ and $\mathbf{B}$, respectively. By Proposition 1,

$$\beta_1 = \frac{n}{3}2^{-(2k+2)} = \frac{1}{2^{k+2}}, \qquad \beta_2 = \beta_3 = \ldots = \beta_{n/3} = 0. \tag{A.1}$$

Then, the inductive hypothesis and Proposition 2 entail that either

$$\begin{cases} \alpha_n + 2\beta_n = \frac{2^{k+1}-3}{2^{k+2}} \\ \alpha_n - \beta_n = \frac{1}{2} \end{cases} \tag{A.2}$$

or

$$\begin{cases} \alpha_n - \beta_n = \frac{2^{k+1}-3}{2^{k+2}} \\ \alpha_n + 2\beta_n = \frac{1}{2} \end{cases} \tag{A.3}$$

is satisfied. System (A.2) has no solution; on the other hand, (A.3) yields

$$\begin{cases} \alpha_n - \frac{1}{2^{k+2}} = \frac{2^{k+1}-3}{2^{k+2}} \\ \alpha_n + 2\frac{1}{2^{k+2}} = \frac{1}{2} \end{cases}$$

which is satisfied by

$$\alpha_n = \frac{2^k - 1}{2^{k+1}}, \tag{A.4}$$

as it can be easily verified, corresponding to the eigenvalues $\frac{2^{k+1}-3}{2^{k+2}}$ and $\frac{1}{2}$ with multiplicity 2 and 1 of matrix $2^{-\boldsymbol{\tau}}$, respectively. On the other hand, Proposition 2, together with equations (A.1), entails

$$\alpha_d = \frac{2^d - 3}{2^{d+1}}, \quad \forall d \in [1, k] \tag{A.5}$$

with multiplicity 2^{k-d}. Hence α_n has multiplicity 1, and in fact $\mathbf{A}$ has exactly $1 + \sum_{d=1}^{k} 2^{k-d} = 1 + \sum_{i=0}^{k-1} 2^i = 2^k = \frac{n}{3}$ eigenvalues.

We now prove that the statement is also true for $k + 1$. We denote by T' the UBT

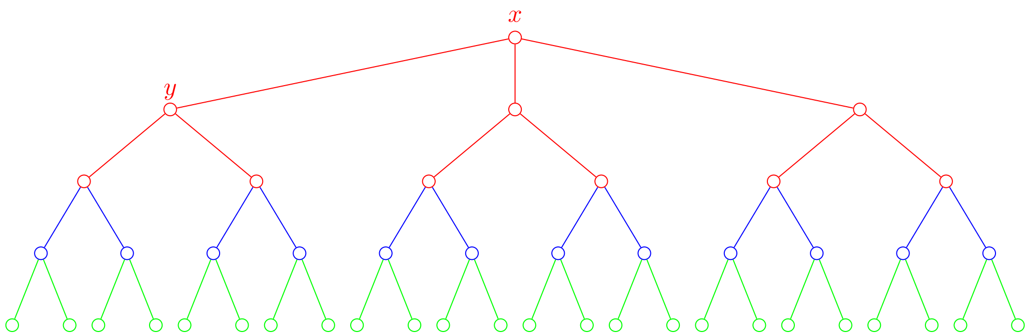

Figure A.2: A representation of a most-balanced UBT with $n = 24$ leaves. Each subtree rooted in the three internal adjacent vertices to x is a rooted binary tree with $\frac{n}{3} = 8$ leaves. In turn, the subtree rooted in y is a rooted binary tree with $\frac{n}{6} = 4$ leaves. The argument can be iterated until we obtain subtrees of size 1 (the leaves of the tree itself).

with $n' = 3 \cdot 2^{k+1}$ taxa. We claim that T' is the realization of the following matrix:

$$\left(\begin{array}{cc|cc|cc} \mathbf{A} & \mathbf{B} & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' \\ \mathbf{B} & \mathbf{A} & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' \\ \hline \mathbf{C}' & \mathbf{C}' & \mathbf{A} & \mathbf{B} & \mathbf{C}' & \mathbf{C}' \\ \mathbf{C}' & \mathbf{C}' & \mathbf{B} & \mathbf{A} & \mathbf{C}' & \mathbf{C}' \\ \hline \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{A} & \mathbf{B} \\ \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{C}' & \mathbf{B} & \mathbf{A} \end{array}\right), \tag{A.6}$$

with $\mathbf{C}' = 2^{-(2k+4)}\mathbf{1}\mathbf{1}^\top$ (see Figure A.2 for visualization). Indeed, there exists a vertex in T and T' such that the removal of such vertex and all its incident edges yields a forest F and F', respectively, consisting of three distinct but isomorphic rooted binary trees. Let us denote such a vertex in T' as x and, without loss of generality, one of the vertices adjacent to x as y. Then, removing y and all its incident edges yields a forest consisting of a connecting component containing x, and two isomorphic connected components; each of such two components is a binary tree isomorphic with any of the binary trees in F. Hence, the distance matrix for the leaves in any of the three connected components of F' is given by

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix}, \tag{A.7}$$

and the claim follows.

As $\mathbf{A}$ and $\mathbf{B}$ are symmetric and commuting, Proposition 2 ensures that matrix (A.7) has eigenvalues

- $\frac{2^{k+1}-1}{2^{k+2}} = \alpha_n + \beta_n$ with multiplicity 1;
- $\frac{2^{k+1}-3}{2^{k+2}} = \alpha_n - \beta_n$ with multiplicity 1;
- $\frac{2^{d}-3}{2^{d+1}} = \alpha_d$, $\forall\ d \in [1, k]$, with multiplicity 2^{k-d+1}.

Now, by Proposition 1,

$$\begin{pmatrix} \mathbf{C}' & \mathbf{C}' \\ \mathbf{C}' & \mathbf{C}' \end{pmatrix} \tag{A.8}$$

has 1 eigenvalue equal to $\frac{n'}{3}2^{-2k+4} = \frac{1}{2^{k+3}}$ and $\frac{n'}{3} - 1$ eigenvalues equal to 0.

Hence, by applying again Proposition 2 on matrix (A.6) and recalling the eigenvalues of matrix (A.7) and (A.8), we obtain that matrix (A.6) has eigenvalues

- $\frac{2^{k+1}-1}{2^{k+2}} + 2\frac{1}{2^{k+3}} = \frac{1}{2}$ with multiplicity 1;

- $\frac{2^{k+1}-1}{2^{k+2}} - \frac{1}{2^{k+3}} = \frac{2^{k+2}-3}{2^{k+3}}$ with multiplicity 2;
- $\frac{2^{k+1}-3}{2^{k+2}}$ with multiplicity 3;
- $\frac{2^d-3}{2^{d+1}}, \forall\, d \in [1, k]$, with multiplicity $3 \cdot 2^{k-d+1}$;

proving the statement. $\square$

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